# Multi-Asset Trading with Quantitative Agents

Research Design, Experimental Development and Out-of-Sample Evaluation

**Final Project • Group 9**

Supplementary Technical Research Report

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## Executive Summary

**Research question.** Can macroeconomic information, risk controls and specialized quantitative agents improve the risk–return trade-off relative to simple ETF portfolios? This project explored this hypothesis through successive research stages: Treasury duration allocation; multi-asset benchmarks and regime-based strategies; a technical, macro and machine-learning agent ensemble; and a final Core–Satellite candidate. (G9-FINAL)

**Final candidate.** The selected *Core35–Satellite65* architecture combines a nominal 35% SPY core with a 65% dynamically selected satellite. Its directional decisions derive from specialized agents and a Genetic-Algorithm-based Orchestrator. Model artifacts and selection parameters were frozen using information through 31 December 2021. The archived final test covers decisions from 7 January 2022 through 9 October 2026, with subsequent next-open execution where available. (G9-FINAL)

**Main empirical result.** On the final archived evaluation calendar, the selected candidate did not outperform either passive benchmark on CAGR, Sharpe ratio or maximum drawdown. The exact values below are read from the archived comparison metrics file; percentages are rounded only for presentation. (G9-OOS)

Table ES.1: *Frozen final evaluation: selected candidate and benchmarks.*

| Strategy | CAGR (%) | Vol. (%) | Sharpe | MDD (%) |
|---|---|---|---|---|
| **Core35–Satellite65** | 5.872 | 11.566 | 0.551 | −26.543 |
| SPY Buy & Hold | 13.116 | 17.294 | 0.799 | −23.424 |
| 60/40 SPY–IEF | 7.388 | 11.174 | 0.694 | −19.840 |

**Interpretation.** Development performance was in-sample with respect to selection of the final candidate. The held-out comparison supplies an informative negative result rather than evidence of robust alpha. The final evaluation followed the frozen inference settings, after-close decisions, next-open portfolio execution and a common daily accounting framework. The data-alignment and information-timing conventions are described in Section 2. (G9-FINAL; G9-OOS)

**Supplementary diagnostics (not a new OOS).** The unchanged OOS ledger, read only for 3 January 2023–9 October 2026, shows Core–Satellite CAGR **14.343%**, Sharpe 1.234 and maximum drawdown −10.035%, compared with 14.061% CAGR for 60/40. The full 2022-start test remains the controlling result. A fixed-target replay that removes only commission and slippage increases the full-OOS Core CAGR from **5.872% to 6.750%**; short-borrow charges remain. An exploratory SMA200 regime filter reaches 9.805% CAGR retrospectively over 2022–2026, but was not validated on a separately protected sample. These analyses explain sensitivity; they do not replace the official result or establish alpha.

*Reader's guide.* Sections 1–13 cover the historical research, final OOS, supplementary diagnostics and reproducible methods; the appendices preserve the experiment register, frozen parameters and source provenance.

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# 1 Research Question and Research Design

## 1.1 Research objective and falsifiable propositions

**Main research question.** Can macroeconomic information, risk controls and specialized quantitative agents improve the *net risk–return trade-off* relative to simple ETF portfolios? The project tested this comparison incrementally, beginning with Treasury duration and extending to multi-asset allocation. The propositions below are research hypotheses, not formally accepted statistical hypotheses. (G9-AB; G9-FINAL)

**H1. Risk conditioning:** an ex-ante volatility-control mechanism may improve risk characteristics compared with static long-only allocations, without requiring directional forecasting.

**H2. Regime dependence:** macro-financial conditions and recent asset momentum may help condition ETF allocation across equities, bonds and gold.

**H3. Complementary signals:** an ensemble of technical, macroeconomic and machine-learning agents may add information beyond stand-alone allocation rules.

**H4. Implementable portfolio design:** a persistent equity core combined with a selectively active satellite may improve participation while retaining the ability to adjust direction and asset choice.

The hypotheses concern different dimensions of performance, not simultaneous improvement in every metric. In the final 2010–2021 development comparison, 60/40 SPY–IEF had a higher Sharpe ratio than Core–Satellite (1.149 versus 0.929). Selection of the candidate was therefore *not* an argmax of development Sharpe; it reflected a documented participation and allocation trade-off. (G9-FINAL)

## 1.2 Investable universe and economic motivation

The first research stage concentrated on Treasury term exposure through **SHY**, **IEF** and **TLT**. Different durations respond differently to yield changes, making duration switching a plausible, measurable mechanism for testing yield and macroeconomic signals. Its archived results motivated a change in research direction rather than a conclusion that Treasury timing could never work. (G9-AB)

The subsequent multi-asset universe comprised four exchange-traded funds: **SPY** (US large-cap equities), **IEF** (intermediate Treasuries), **GLD** (gold), and **SHY** (short-duration Treasuries). These instruments offer interpretable exposures to equity risk, duration risk, a gold allocation and a relatively short-duration defensive alternative. The eligible universe is limited; neither holding four possible instruments nor operating an agent ensemble automatically implies diversified *realized* portfolio risk. (G9-FINAL)

## 1.3 An incremental and decision-driven research program

Figure 1.1 summarizes the research sequence. Earlier specifications remain documented, including those simplified or abandoned after unfavorable results.

## 1.4 Chronology, information sets and comparison rules

The distinct calendars appear in Table 2.3. Treasury diagnostics used weekly shifted close-to-close returns; final portfolios used weekly post-close decisions, next-open execution and daily NAV. Model-estimation dates and common performance dates need not coincide. The 2017–2021 validation had been examined

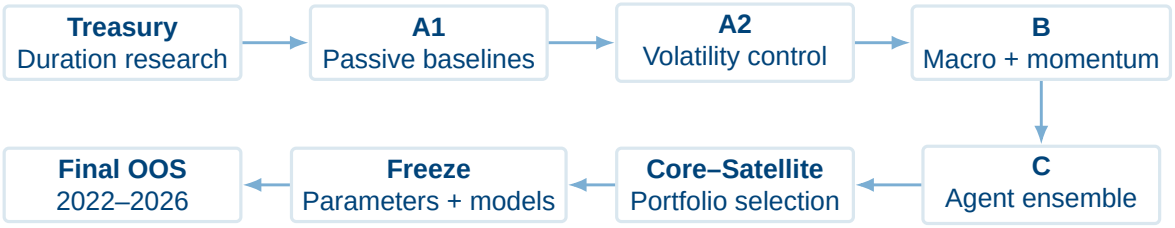



Figure 1.1: *Research sequence: from a Treasury-specific hypothesis to an unchanged final multi-asset candidate. The arrows show the order of investigation, not proof that each added layer improved performance.*

historically, and final strategy selection used 2010–2021 development data. RF/SVM out-of-fold forecasts do not make the GA-selected ensemble independently OOS. (G9-AB; G9-FINAL)

### 1.5 Evaluation criteria and study scope

Final benchmarks are SPY Buy & Hold and weekly-rebalanced 60/40 SPY–IEF. Comparisons use the same performance dates and daily accounting assumptions, while allowing each policy to retain its own weights. We report CAGR, realized volatility, Sharpe (zero risk-free rate, 252 trading days), maximum drawdown and implementation costs. The frozen candidate did not outperform either benchmark on CAGR, Sharpe or drawdown; Section 8 provides the full evidence. (G9-FINAL; G9-OOS)

## 2 Data Sources, Feature Engineering and Temporal Protocol

### 2.1 Data sources and investable instruments

The study uses locally archived ETF prices from Yahoo Finance (via `yfinance`, with automatic price adjustment) and macroeconomic series from FRED. Provider documentation describes the collection interface; the analysis itself uses the locally archived, frozen CSV snapshots rather than live downloads. (yfinance contributors 2026; Federal Reserve Bank of St. Louis 2026) Table 2.1 separates the four tradable final ETFs from TLT, used only in historical Treasury research. (G9-DATA; G9-CODE)

Table 2.1: *Archived data inputs and analytical roles.*

| **Input family** | Archived files / fields | Role in the research |
|---|---|---|
| **Tradable ETFs** | `SPY`, `IEF`, `GLD`, `SHY`; daily adjusted Open, Close, High, Low and Volume | Equity, intermediate-duration Treasury, gold and short-duration Treasury exposures. |
| **Treasury archive** | `TLT.csv` and Treasury yield histories | Earlier duration-timing experiments; TLT is not a final Satellite candidate. |
| **Nominal rates** | `DGS3MO`, `DGS2`, `DGS5`, `DGS10`, `DGS30`, `FEDFUNDS` | Treasury levels, yield-curve slopes, changes and policy-rate conditions. |
| **Inflation / breakevens** | `CPIAUCSL`, `CPILFESL`, `T5YIE`, `T10YIE` | Inflation dynamics and market-implied inflation components. |
| **Market stress** | `VIXCLS` | Level and changes in equity-implied volatility. |

The archived raw prices begin in January 2004 for SPY, IEF, SHY and TLT, and on **18 November 2004** for GLD. Thus 18 November 2004 is the first common date in the four-asset price panel; older observations for individual ETFs may be used for standalone historical diagnostics but not as a complete four-asset trading

panel. The raw histories extend beyond some evaluation cutoffs and are always restricted to the period authorized for the specific calculation. (G9-DATA; G9-CODE)

## 2.2 Data cleaning, price adjustment and common sessions

Market ingestion is deliberately strict. The reader converts the Date field to a chronological index and checks for duplicate or invalid dates, numerical price validity and required Open/Close fields. For a joint portfolio backtest, `load_daily_market_data` restricts the data to the *intersection of actual trading sessions* in the four ETFs. It records missing sessions and requires at least 99% agreement between the intersection and union over the common date span. Importantly, **tradable prices are not forward-filled** to manufacture missing Open or Close prices. (G9-CODE)

A saved development-only panel, `data/processed/market_daily.csv`, contains **4,310 common daily rows** from 18 November 2004 to 31 December 2021 and eight adjusted price columns (Open/Close for each of the four ETFs). The frozen final test instead loads the archived raw source histories and reconstructs its permitted calendar; the 2022+ input audit records **1,197 common sessions** from 3 January 2022 through 9 October 2026. These are different objects, not interchangeable sample sizes. (G9-DATA; G9-AUDIT)

## 2.3 Weekly decisions and information-to-execution timing

The investment policy runs at a weekly decision frequency, but positions and portfolio value are accounted for at daily frequency. A critical implementation detail is that a weekly decision is timestamped with the *last actual trading session of that week*, not mechanically assigned to a Friday that might have been a market holiday. The function `build_weekly_decision_dates` derives these dates from the session calendar; a Thursday close is therefore a legitimate weekly decision when Friday is closed. (G9-CODE)

Figure 2.1 shows the execution convention. Signal construction may use the closing observation at $t$ because the trade occurs only at the next available opening session $t^+$. In particular, the backtester does not trade at the same Close that provided the signal input.




Figure 2.1: *Weekly post-close decision and next-open execution. Performance accounting continues on the actual daily market calendar.*

Earlier weekly `W-FRI` research panels should not be mistaken for the final actual-session decision calendar. (G9-CODE)

## 2.4 Engineered market and macro-financial features

The feature layer is centralized in `src/features.py`. Its market decision panel retains adjusted Close at each weekly decision, a realized return since the preceding decision, and momentum over 21 and 63 actual trading sessions. A separate asset-level ML feature constructor supplies one-week return; 4-, 13-, 26- and 52-week momentum; a 13-week rolling return standard deviation; distance from the 52-week high; and the other ETFs' 13-week momentum. These are *backward-looking measurements at a decision date*, not forward targets. RF and SVM prediction targets and horizons will be described in the agent methodology chapter. (G9-CODE)

For macroeconomic information, `src/data_loader.py` combines rate, inflation and market-stress his-

Table 2.2: *Feature construction and observation frequency.*

| **Feature block** | Representative transformations | Timing convention |
|---|---|---|
| **ETF decision features** | Weekly Close and return; 21-/63-session price momentum. | Calculated at the actual weekly Close. |
| **Asset-level ML** | Momentum at 4/13/26/52 weeks; volatility at 13 weeks; 52-week high distance; cross-ETF momentum. | Backward rolling windows evaluated on the weekly panel. |
| **Yield curve** | 10Y–2Y, 10Y–3M, 30Y–10Y spreads; recent changes. | Macro observation series aligned backward to decisions. |
| **Rates / inflation** | Treasury-yield changes; policy rate; year-over-year CPI; breakevens. | Series-specific lag/availability convention; no forward-dated joins. |
| **Market stress** | VIX level and approximately four-week changes. | Observations incorporated into the decision-date macro feature panel. |

tories; derives changes, spreads and normalized quantities; and aligns feature rows to decision dates with a *backward as-of join*. The archived configuration uses conservative release-date offsets of **45 calendar days** for monthly CPI and core CPI and **35 calendar days** for the policy-rate observation; ordinary daily series are aligned using the most recent eligible observation. The codebase also includes an event-timestamp alignment function that joins on a time-zone-aware `available_at` field whenever that release metadata is supplied. These are distinct data-handling mechanisms, both designed to keep the decision information set chronologically ordered. (G9-CODE)

## 2.5 Research windows, freeze boundary and final test calendar

The project contains multiple valid calendars with different analytical purposes. The initial A/B training specification began on 20 May 2005 and originally terminated in December 2016, while 2017–2021 was used for historical validation and later development. The final Core–Satellite performance comparison uses a **common 2010–2021 development trading calendar**; model estimation may use earlier observations or different rolling windows. The final selected candidate was frozen using information through **31 December 2021**. (G9-AB; G9-FINAL)

Table 2.3: *Distinct date ranges and correct interpretation of reported sample counts.*

| Object / window | Period or count | Interpretation |
|---|---|---|
| Development market panel | 18 Nov 2004–31 Dec 2021; 4,310 rows | Historical daily price data, including warm-up. |
| Initial A/B model research | 20 May 2005–30 Dec 2016 | Original training era; not the final common return window. |
| Final candidate development | 2010–2021 | Strategy-selection comparison: in-sample for the chosen candidate. |
| Common raw OOS sessions | 3 Jan 2022–9 Oct 2026; 1,197 sessions | Available market records before the first OOS decision. |
| Final weekly decision series | 7 Jan 2022–9 Oct 2026; 249 decisions | Agent and Satellite decisions; not an order count. |
| Final daily NAV series | 7 Jan 2022–9 Oct 2026; 1,193 rows | Actual accounting calendar; first execution on 10 Jan 2022. |

The final decision on 9 October 2026 does not have a later executable opening session within the archived test and is therefore not an executed new order. Likewise, the number of weekly decisions, the number of days with nonzero trades and cumulative turnover measure different things. (G9-FINAL; G9-AUDIT)

### 2.6 Source integrity and reproducible inputs

The archived release records SHA-256 hashes, checks chronology and protects frozen outputs against accidental overwrite. The full replay procedure is given in Section 11. (G9-AUDIT; G9-CODE)

## 3 Treasury Research: Macro Regimes and Duration Timing

### 3.1 Economic rationale and initial design

Initial research asked whether rates, inflation and yield-curve information could guide Treasury duration across SHY, IEF and TLT. Term-structure research motivates testing such predictors (Campbell and Shiller 1991; Cochrane and Piazzesi 2005), but evidence on longer-horizon yields and bond risk premia does not establish predictability of *next-week ETF returns*. (G9-TREAS)

The archived study (8–23 October 2026) used weekly adjusted-price observations and a *shifted-weight close-to-close* approximation, with 3 bps charged per turnover unit; it did not use the later daily next-open engine. Its heuristic score combined the backward 13-week yield change (2Y, later 10Y/30Y), CPI year-on-year trends, the 10Y–2Y curve state and rolling 10Y yield volatility relative to an expanding median. The features condition duration decisions, not identified latent economic states. (G9-TREAS)

The initial deterministic specification combined these inputs into a heuristic regime score. The yield-direction component received a larger weight (absolute weight 2) than the inflation and curve indicators (unit weights), and the original decision boundaries were $-2$ and $+2$. The resulting regimes mapped to long TLT, long IEF, or a defensive configuration with $+50\%$ SHY / $-50\%$ TLT. These were *designed allocation rules*, not parameters estimated through a statistical identification of latent macroeconomic regimes. Sixteen binary configurations were theoretically possible, of which 13 appeared in the archived development panel. (G9-TREAS)

### 3.2 Baseline results and threshold sensitivity

The first evaluation tested whether the original score improved upon simply holding the ETFs. Table 3.1 reports archived *cumulative* returns, not compound annual growth rates. The underlying active-strategy evaluation covered 606 training weeks (27 May 2005–30 December 2016) and 261 historical-validation weeks (6 January 2017–31 December 2021). The passive comparisons refer to those same respective windows. (G9-TREAS)

Table 3.1: *Archived Treasury rules versus passive alternatives: cumulative return (%).*

| Strategy | Train 2005–2016 | Validation 2017–2021 |
|---|---|---|
| Original regime score: 2Y proxy; thresholds $\pm 2$ | −3.11 | 2.72 |
| Exploratory score thresholds $\pm 3$ | 34.55 | 5.32 |
| SHY Buy & Hold | 27.94 | 7.58 |
| IEF Buy & Hold | 72.03 | 18.98 |
| TLT Buy & Hold | 93.50 | 38.18 |
| Equal-weight true Buy & Hold | 64.49 | 21.58 |

The original score underperformed passive comparators. Expanding thresholds to $\pm 3$ improved training performance but not validation competitiveness; it was a subsequent research modification, not an independently validated selection. (G9-TREAS)

## 3.3 Ablations: yield maturity, defensive positions and score meaning

Train-only ablations tested 2Y, 10Y and 30Y yield-direction proxies and then the defensive leg (Table 3.2). The 30Y proxy and long-only SHY defense gave the better observed results within these comparisons. (G9-TREAS)

Table 3.2: *Archived train-only duration experiments: rates proxy and defensive allocation.*

| Train-only specification | Cum. return (%) | Sharpe | MDD (%) |
|---|---|---|---|
| **13-week yield-direction proxy; original defensive allocation** | | | |
| 2Y Treasury yield | −3.11 | −0.102 | −36.68 |
| 10Y Treasury yield | 19.20 | 0.084 | −25.21 |
| 30Y Treasury yield | 46.96 | 0.267 | −19.89 |
| **30Y proxy; defensive allocation varied** | | | |
| $+50\%$ SHY, $-50\%$ TLT | 46.96 | 0.267 | −19.89 |
| $+100\%$ SHY | 58.20 | 0.340 | −19.89 |
| $-100\%$ TLT | 35.10 | 0.184 | −22.22 |

The 30Y proxy has a defensible economic interpretation because long-duration TLT is more sensitive to long-term yield movements. Nevertheless, the comparison was used to develop the specification; the three defensive portfolios also have materially different interest-rate, short-selling and implementation risk. *Nominal dollars are not duration-matched risk units*. These alternatives were retained as research diagnostics rather than presented as equally risky investment substitutes. (G9-TREAS)

A separate score audit directly examined the next-week TLT-minus-SHY return spread within each original score category. Under the 2Y proxy, the conditional mean spread was $+14.27$ bps in the defensive category, $+18.68$ bps in the intermediate category and $-16.25$ bps in the intended long-duration category. Replacing the rate proxy with 30Y gave respectively $+1.19$, $+13.83$ and $+6.32$ bps. Neither version produced the clean monotonic ranking one would hope to observe from a score designed to rank favorable duration conditions. These are descriptive conditional means, not causal estimates. (G9-TREAS)

## 3.4 Predictive experiment: expanding-window Ridge regression

Ridge forecast next-week $r_{\text{TLT}} - r_{\text{SHY}}$ from the 13-week 30Y yield change, 10Y–2Y spread, CPI YoY and 13-week 10Y yield-change volatility. An expanding-window pipeline standardized predictors within each fit, used fixed $\alpha = 10$, and compared with the expanding historical mean. (G9-TREAS)

The archived diagnostic contained 619 complete feature/target rows, an initial 156-row estimation sample and **463 expanding-window predictions** dated 15 February 2008–23 December 2016. It was an internal *train-only* forecast evaluation, not the final OOS strategy evaluation. The comparison was unfavorable to Ridge on all three reported metrics (Table 3.3). (G9-TREAS)

Table 3.3: *Train-only next-week TLT–SHY forecasting diagnostic (463 observations).*

| Diagnostic | Ridge | Expanding mean |
|---|---|---|
| RMSE (bps) | 200.360 | 199.103 |
| MAE (bps) | 154.316 | 151.931 |
| Directional accuracy (%) | 50.54 | 54.43 |

The relative predictive $R^2$ was $-0.01267$, equivalent to a mean-squared error approximately $1.27\%$ higher

than the expanding-mean reference. The experiment does *not* show that rates returns are inherently unforecastable; it establishes only that this small feature set, weekly target and regularized linear model did not outperform the chosen naive predictor. Accordingly, Ridge was not promoted into a Treasury trading strategy. (G9-TREAS)

## 3.5 The 30Y yield-timing rule and historical validation

The final prominent Treasury diagnostic removed the multi-component score entirely. It implemented a two-state rule based on the 30Y Treasury yield over the preceding 13 weeks:

$$w_{t+1} = \begin{cases} 100\%\ \text{TLT}, & y_{30,t} - y_{30,t-13\text{w}} < 0, \\ 100\%\ \text{SHY}, & \text{otherwise}. \end{cases} \tag{3.1}$$

The weight $w_{t+1}$ denotes the *subsequent* weekly holding, using the historical shifted close-to-close execution approximation and the same 3-bps turnover-cost convention. Because TLT is duration-sensitive and SHY is relatively defensive, the rule is economically transparent; it is not a claim of risk-matched comparison. (G9-TREAS)

Table 3.4: *30Y yield timing: archived weekly cumulative returns and implementation diagnostics.*

| Historical window | Weeks | Timing (%) | TLT B&H (%) | Vol. (%) | MDD (%) | Turnover |
|---|---|---|---|---|---|---|
| Train: 15 Feb 2008–23 Dec 2016 | 463 | 91.80 | 73.19 | 11.51 | −16.47 | 101 |
| Validation: 6 Jan 2017–31 Dec 2021 | 261 | 7.15 | 38.18 | 10.95 | −13.76 | 94 |

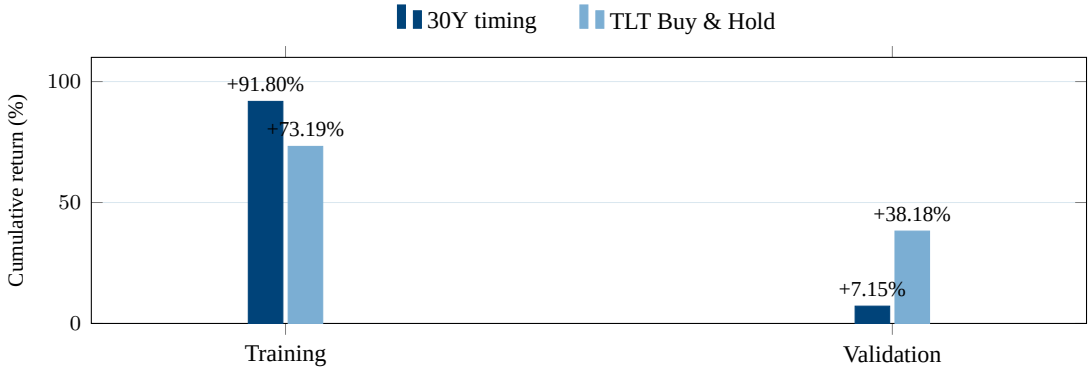



Figure 3.1: *The same timing rule's advantage in the exploratory training diagnostic did not carry over to the later historical validation. The bars are* separate cumulative-return windows*; they are not annualized growth rates.*

The attractive +91.80% training return did not persist in validation (+7.15%, versus +38.18% for TLT). The rule also lagged SHY, IEF and equal-weight buy-and-hold in that period. This is evidence about the *tested specification and dates*, not every Treasury timing strategy. (G9-TREAS)

## 3.6 Research decision and transfer to the multi-asset program

The team **paused and archived** Treasury-specific tuning after negative comparative evidence, retaining its research history. The subsequent SPY/IEF/GLD/SHY universe tested whether mixing equity beta, duration and gold with risk management and agents would offer a better allocation problem. All Treasury tables

in this chapter report historically archived weekly cumulative-return diagnostics, *not* final daily-engine CAGR. (G9-TREAS; G9-AB)

# 4 Multi-Asset Research: Baselines, Risk Control and Regime Allocation

## 4.1 The pivot: a broader but still interpretable investment universe

The four-ETF universe SPY/IEF/GLD/SHY was selected to test whether incremental complexity improved an interpretable portfolio. A1 establishes passive baselines; A2 controls estimated volatility without forecasting direction; B adds macro budgets and momentum. Comparisons use weekly post-close decisions, next-open execution, daily NAV, 2 bps commission plus 1 bp slippage and, for shorts, 1% annualized borrow fees. This differs from the earlier Treasury engine. (G9-AB; G9-LOG)

## 4.2 Level A1: the baselines against which complexity is judged

The four A1 specifications deliberately span distinct reference portfolios. **SPY Buy & Hold** represents uninterrupted equity beta; **60/40 SPY–IEF** is a weekly-rebalanced stock–bond benchmark; **Equal Weight** rebalances SPY/IEF/GLD/SHY to 25% each week; and **Equal Weight Buy & Hold** starts at those proportions but subsequently allows the weights to drift. The two equal-weight versions are therefore not interchangeable. Buy-and-hold policies trade at initial implementation rather than continuously restoring starting weights. (G9-AB; G9-LOG)

A1 isolates the role of allocation, rebalancing and equity participation before introducing active rules.

## 4.3 Level A2: controlling ex-ante volatility without a direction forecast

A2 is motivated by the distinction between managing portfolio volatility and forecasting returns; volatility-managed factor strategies have been studied separately in the literature (Moreira and Muir 2017), but the rule below is a different, ETF-specific implementation. It remains **long-only and fully invested** and starts from the four-asset equal-weight reference $w^{\mathrm{EW}} = (0.25, 0.25, 0.25, 0.25)^\top$. At each decision, the covariance matrix $\widehat{\Sigma}_t$ is the sample covariance of the latest **126 daily return observations**, annualized by 252. If the estimated volatility of equal weight does not exceed **8% annually**, the reference weights are maintained. Otherwise, A2 solves the constrained, convex distance-minimization problem

$$\begin{aligned} \underset{w}{\text{minimize}} \quad & \|w - w^{\mathrm{EW}}\|_2^2, \\ \text{subject to} \quad & \mathbf{1}^\top w = 1, \qquad w_i \geq 0, \\ & \sqrt{w^\top \widehat{\Sigma}_t w} \leq 0.08. \end{aligned} \tag{4.1}$$

Implemented using CVXPY/CLARABEL, A2 falls back to the fully invested minimum-variance portfolio if the 8% bound is infeasible. The target is an *estimated ex-ante ceiling where feasible*, not guaranteed realized risk or directional forecasting. Optimization was invoked on 87 of 607 training decisions. (G9-AB; G9-LOG; G9-CODE)

## 4.4 Level B: macro-state budgets and price momentum

### 4.4.1 Deterministic macro states and budget design

Level B introduced two observed state dimensions: `VIXCLS` as a market-stress indicator and `T10YIE` as the 10-year Treasury inflation-breakeven rate. A value is classified as *high* when it exceeds the median of its **previous 52 weekly observations**; the historical median used as a threshold is shifted to exclude the current observation. The first label in `low_high` or `high_low` describes *VIX stress*; the second describes the inflation-breakeven state. These are deterministic rules rather than fitted latent states. (G9-AB; G9-CODE)

Table 4.1: *Predefined Level B allocation budgets by deterministic macro state (percent of total portfolio).*

| Regime (VIX / breakeven) | SPY | IEF | GLD | SHY |
|---|---|---|---|---|
| low_low | 25 | 30 | 20 | 25 |
| low_high | 25 | 20 | 30 | 25 |
| high_low | 15 | 30 | 20 | 35 |
| high_high | 15 | 20 | 30 | 35 |
| Equal-state average | 20 | 25 | 25 | 30 |

The budgets decrease SPY and raise SHY under high VIX, and favor GLD over IEF under high breakevens. These are heuristic *pre-filter budgets*, not optimized or necessarily fully invested weights. (G9-AB)

### 4.4.2 B_v1: signed momentum with an additional directional gate

Time-series momentum has been documented across futures markets at longer horizons (Moskowitz, Ooi, and Pedersen 2012); this does not establish an edge for the present ETF thresholds. For each ETF $i$ at weekly decision $t$, Level B measures daily-price momentum over 21 and 63 sessions:

$$m_{i,h,t} = \frac{P_{i,t}}{P_{i,t-h}} - 1, \qquad h \in \{21, 63\}. \tag{4.2}$$

The original directional signal $s_{i,t}^{(1)}$ is $+1$ if *both* horizons are positive, $-1$ if *both* are negative, and zero if their signs disagree (or neither condition holds). B_v1 additionally suppresses a positive SPY signal in a high-VIX state and a positive IEF signal in a high-breakeven state. With macro-state budget $b_{i,t}$, the initial signed target is $\widetilde{w}_{i,t} = b_{i,t} s_{i,t}^{(1)}$. (G9-AB; G9-CODE)

The resulting portfolio can hold *long, short and flat* positions. A proportional volatility scaling factor is calculated from the 126-day covariance estimate, never leverages a portfolio upward, and keeps the aggregate gross target exposure at or below 100% of NAV. The backtester models daily quantities, cash, transaction costs, 1% annualized borrow fees for shorts and simplified collateral/margin constraints. The nominally symmetric $\{-1, 0, +1\}$ signal alphabet therefore does *not* imply symmetric economic risk across ETF instruments. (G9-AB; G9-CODE)

### 4.4.3 B_v2: the Long/Flat specification selected after ablation

The training ablation indicated that short positions and the supplementary directional macro gate penalized the tested B_v1 implementation. B_v2 therefore preserves the four macro *budgets* but drops the macro *directional gate* and maps momentum into a strict long/flat signal:

$$s_{i,t}^{(2)} = \begin{cases} 1 & \text{if } m_{i,21,t} > 0 \text{ and } m_{i,63,t} > 0, \\ 0 & \text{otherwise.} \end{cases} \qquad \widetilde{w}_{i,t}^{(2)} = b_{i,t} s_{i,t}^{(2)}. \tag{4.3}$$

A single estimated-volatility cap is applied to the *actual nonnegative weight vector after this long/flat conversion*. Writing the annualized volatility of the raw portfolio as $\widehat{\sigma}_t(\widetilde{w})$, the scaling factor when volatility is nonzero is

$$\kappa_t = \min\left\{1, \frac{0.08}{\widehat{\sigma}_t(\widetilde{w})}\right\}, \qquad w_{i,t} = \kappa_t \widetilde{w}_{i,t}. \tag{4.4}$$

Unused investment capacity is retained as *non-interest-bearing cash*; B_v2 does not lever up to fill an unused volatility budget. It is thus **Long/Flat rather than Long/Short, market-neutral, or fully invested**. It is not equivalent to taking B_v1's already capped weights and replacing negative positions with zero, because the cap must be recalculated after the signal-policy change. (G9-AB; G9-LOG; G9-CODE)

## 4.5 Training performance, 2005–2016: risk control versus participation

Table 4.2 records the historical training evaluation from **20 May 2005 through 30 December 2016**. These are *annualized CAGR, realized volatility and Sharpe ratios, plus maximum drawdown*, all produced by the common daily multi-asset engine and net of its modeled trading costs. They should not be conflated with the Treasury *cumulative* returns or with the narrower 2010–2016 comparison below. (G9-AB)

Table 4.2: *A1/A2/B performance in the original 2005–2016 training comparison.*

| Strategy | CAGR (%) | Vol. (%) | Sharpe | MDD (%) |
|---|---|---|---|---|
| SPY Buy & Hold | 7.77 | 19.68 | 0.479 | −55.19 |
| 60/40 SPY–IEF weekly | 7.13 | 10.88 | 0.688 | −32.00 |
| Equal Weight weekly | 6.54 | 6.99 | 0.942 | −15.69 |
| Equal Weight Buy & Hold | 6.16 | 8.75 | 0.727 | −17.63 |
| **A2: volatility control** | 6.24 | 6.41 | 0.977 | −14.28 |
| B_v1: Long/Short | 0.24 | 5.55 | 0.071 | −17.55 |
| **B_v2: Long/Flat** | 2.64 | 4.17 | 0.646 | −7.01 |

A2 improved training Sharpe over weekly equal weight (0.977 versus 0.942) and reduced drawdown, although CAGR was slightly lower. B_v1 grew little; B_v2 improved on B_v1 after simplification but still lagged A2 on CAGR and Sharpe. These are risk-control and participation comparisons, not proof of predictive alpha. (G9-AB; G9-LOG)

The B_v1 internal attribution recorded approximately **USD 31,900** of long-position gains offset by **USD 20,300** of short-position losses before costs, followed by roughly **USD 6,580** in trading costs and **USD 2,170** in borrowing charges. These rounded, diagnostic components are not a substitute for a complete causal decomposition. B_v2 had **54.18% mean target exposure** in this training evaluation, which helps explain why the much smaller realized volatility coexisted with modest returns. This target exposure statistic should not be mixed with the *realized gross exposure* measured on the later common 2010–2016 NAV calendar. (G9-AB)

## 4.6 Ablation evidence and the decision to retain B_v2

Table 4.3 isolates the tested Level B rule families, without assuming an unrecorded exhaustive factorial study.

The available budget comparison further quantifies the last point. On the original training calendar, B_v2's **Sharpe 0.6464** compares with **0.6094** using the fixed equal-state-average budget vector $(20, 25, 25, 30)\%$, and **0.5955** using fixed 25%-per-ETF budgets, under the same long/flat volatility-cap structure. The improvement over the equal-state-average comparator is only **0.0370 Sharpe units**. This is an observed

Table 4.3: *Level B train-only ablation: intervention, observed effect and research decision.*

| Component tested | Training observation | Decision |
|---|---|---|
| Signed short exposure | Short leg lost money in the examined B_v1 attribution; removing shorts improved development Sharpe. | Remove in B_v2. |
| Additional directional macro gate | In the inspected long/flat configuration, removing this extra gate improved the result. | Remove the gate. |
| Two-horizon momentum | Long eligibility required positive 21- and 63-session momentum simultaneously. | Retain. |
| Macro-conditioned budgets | A small Sharpe advantage over fixed budgets was observed under the B_v2 risk cap. | Retain budgets; no proof of standalone alpha. |
| Risk scaling after signal mapping | Scaling the long/flat portfolio after dropping shorts materially determines effective risk and cash. | Apply cap after mapping. |

training difference, not an independently established effect of regime timing; realized exposures and trading paths differ. (G9-AB; G9-LOG)

B_v1 was archived; the simpler B_v2 was retained for Level C comparison, not because it maximized Sharpe across A/B. (G9-LOG)

## 4.7 Matched 2010–2016 comparison and exposure interpretation

When Level C was integrated, the A/B strategies were rerun on a **common 2010–2016 performance calendar**, using identical simulated initial capital (USD 100,000), market dates, next-open convention, daily NAV accounting and cost assumptions. This comparison covers **365 weekly decisions**, not the 607 decision dates of the broader 2005–2016 A/B experiment. The complete nine-strategy comparison, including both Level C variants, is retained in Table D.1. The seven A/B Sharpe values are plotted in Figure 4.1; the next chapter discusses C_v1 and C_v2. (G9-C)

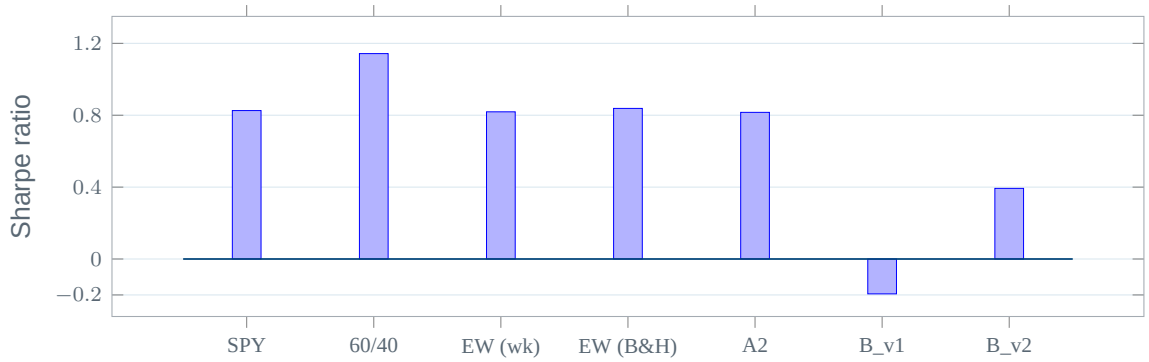



Figure 4.1: *The development Sharpe comparison on identical 2010–2016 performance dates. The lower-risk B variants did not improve risk-adjusted returns relative to the simpler A-level portfolios.*

The common-calendar comparison makes the low-participation mechanism explicit. B_v2's mean *realized gross* exposure was **51.939%** in 2010–2016, versus approximately 100% for the A1 and A2 portfolios. Its small drawdown and low volatility therefore should not automatically be interpreted as a successful forecast of adverse market conditions. Note that **51.939% realized gross exposure over 2010–2016** and **54.18% mean target exposure over 2005–2016** refer to different definitions and periods; both figures are retained in their original context. (G9-C; G9-AB)

### 4.8 Historical B_v2 validation and handoff to the agent experiments

The historical B_v2 record also includes a separate **2017–2021 validation diagnostic**, run after the B_v2 training specification was selected. This run did not redefine the 2022+ final OOS test and should not be combined arithmetically with the original training or matched 2010–2016 tables. The recorded validation-only results were: **1,259 daily sessions**, cumulative return **+8.919%**, CAGR **1.725%**, volatility **3.373%**, Sharpe **0.5239** and MDD $-4.696\%$. The normalized USD 108,919 ending amount refers to a USD 100,000 valuation base at the start of this diagnostic, not to restarting the actual portfolio's positions. (G9-AB)

The Level B evidence motivates testing whether Level C adds conditional information beyond simple momentum, while preserving the distinct training, validation and development calendars. Its GA-selected ensemble remains in-sample on the common 2010–2016 comparison. (G9-C; G9-FINAL)

## 5 Agent Architecture and Level C

### 5.1 Complementary signals within a common interface

Level C tested whether heterogeneous quantitative agents could improve the low-exposure B_v2 strategy developed in Section 4. The agents are *quantitative signal modules*, not conversational AI systems. Each supplies a weekly direction $s_{j,i,t} \in \{-1, 0, +1\}$ and a model-specific confidence score $c_{j,i,t} \in [0, 1]$ for ETF $i$. Their objectives differ: technical rules describe price configurations; the macro rule translates observable economic conditions; and RF/SVM estimate future direction from engineered features. The RF and SVM model families have distinct statistical foundations (Breiman 2001; Cortes and Vapnik 1995); the implemented training windows and targets are specific to this project. The confidence values are **not mutually calibrated probabilities of trading success**. (G9-C; G9-FINAL; G9-CODE)

Table 5.1: *Level C agent families and their distinct information channels.*

| Agent | Operational signal or target | Confidence basis |
|---|---|---|
| SMA/EMA | Fast EMA versus slower SMA: signed trend, with optional neutral band | Normalized gap between the averages |
| Z-Score | Rolling standardized-price breakout; frozen specification uses momentum mode | Magnitude of active Z-score |
| Macro Yield Curve | Deterministic rates, curve and inflation score mapped to ETF-specific actions | Score magnitude, conditional on activity |
| Random Forest (RF) | Binary sign of the next weekly ETF price return | Distance of $\widehat{P}(\text{up})$ from 0.5 |
| Support Vector Machine (SVM) | Binary direction over the next four weekly decisions | Scaled decision margin, $\tanh(\lvert z \rvert)$ |

SMA/EMA produces a positive signal when the fast exponential average lies above the slower simple average; the Z-Score strategy also allows mean reversion as a configurable alternative, but the *frozen* model uses momentum. The macro agent remains rule-based. The archived development path excluded FinBERT from the final candidate because a suitable, historically timestamped news-sentiment input was not available for that experiment. More detailed feature lists and frozen per-asset settings are reserved for Appendix B, rather than the main narrative. (G9-FINAL; G9-CODE)

## 5.2 Supervised targets, chronological training and model selection

The **RF** labels the sign of the next weekly ETF return. Its historical development protocol uses a rolling **five-year training window followed by a one-year prediction window**; a small hyperparameter grid is scored on the training block, and predictions are then recorded for the subsequent year. Parameter combinations are consolidated across folds before fitting the saved model on eligible data through 2021. RF output probabilities supply a *ranking score*: the subsequent confidence transformation does not establish external probability calibration. (G9-CODE; G9-FINAL)

The **SVM** predicts a different outcome: whether the ETF price is higher **four weekly decisions ahead**. It uses a scaler–SVC pipeline, temporal cross-validation with a **four-week gap** corresponding to the label horizon, and annual forward predictions based only on labels already matured at that year's training boundary. A neutral band around the scaled decision score defines the flat region; the saved classifier is then trained through the 2021 cutoff. These are distinct predictive horizons, so RF and SVM accuracy cannot be compared as though their labels were identical. (G9-CODE; G9-FINAL)

The two supervised modules thus produce historical *out-of-fold / walk-forward base signals*. That does **not** make the complete ensemble independent of the development data: the combination rule and its selection are also learned on historical signals and returns. Level C's matched historical portfolio results below remain *in-sample with respect to the ensemble decision*. (G9-C)

## 5.3 GA Orchestrator: aggregation rather than another return predictor

For agents $j = 1, \ldots, N$, the Orchestrator transforms confidences by agent-specific exponents $\alpha_j$ and forms a bounded consensus score. Ignoring numerical clipping for readability, its calculation is

$$z_{i,t} = \frac{\sum_{j=1}^{N} w_j s_{j,i,t}\, c_{j,i,t}^{\alpha_j}}{\sum_{j=1}^{N} w_j c_{j,i,t}^{\alpha_j} + \varepsilon}, \qquad s_{i,t}^{\text{orch}} = \begin{cases} +1, & z_{i,t} > \theta_L, \\ -1, & z_{i,t} < -\theta_S, \\ 0, & \text{otherwise.} \end{cases} \tag{5.1}$$

where $\sum_j w_j = 1$. The real-coded genetic algorithm searches **agent weights, confidence exponents, and asymmetric long/short thresholds** against an economic fitness measure incorporating weekly trading friction, drawdown and activity/concentration constraints. It is not an additional neural or language model. The GA's weekly proxy fitness should not be confused with the final strategy's separate *daily next-open execution engine*. (G9-CODE; G9-FINAL)

## 5.4 Level C portfolio experiments and observed trade-off

The original variants used the Level B macro-state budgets and estimated-volatility cap, but altered position eligibility. In **C_v1**, a long position required *both* positive B_v2 momentum and positive Orchestrator direction with nonzero confidence; otherwise that ETF stayed flat. **C_v2** also allowed a short when signed two-horizon momentum and Orchestrator direction were both negative. Both applied the same **8% ex-ante volatility cap once, after signed positions were formed**, with target gross exposure at most 100%. These variants are distinct from the subsequently selected Core–Satellite portfolio. (G9-C)

Table 5.2 isolates the relevant results on the **matched 2010–2016 calendar**: 365 weekly decisions, a common USD 100,000 starting NAV, next-open execution, daily valuation and the same stated transaction-cost convention. The table presents *realized gross exposure*, not nominal target weight or the different

2010–2021 development statistics later used to select Core–Satellite.

Table 5.2: *Matched 2010–2016 development: risk–return and realized gross participation.*

| Strategy | CAGR (%) | Vol. (%) | Sharpe | MDD (%) | Gross exp. (%) |
|---|---|---|---|---|---|
| 60/40 weekly | 9.566 | 8.296 | 1.143 | −8.470 | 99.943 |
| B_v2: Long/Flat | 1.366 | 3.621 | 0.393 | −7.011 | 51.939 |
| C_v1: Long/Flat | 1.793 | 3.001 | 0.607 | −3.693 | 46.302 |
| C_v2: Long/Short/Flat | 2.091 | 3.272 | 0.649 | −3.693 | 48.864 |

C_v2 improved on B_v2 within this development comparison, but **neither C variant matched 60/40 Sharpe**. Their lower drawdowns coincided with roughly half the benchmark's realized market exposure. In C_v2, 44 of the 365 weeks included at least one short position. The separate *exposure sensitivity*, which scaled weights before the volatility cap without adding new signals, illustrated the trade-off: at the 12% cap, increasing the C_v2 scale from $1\times$ to $2\times$ raised CAGR from **2.092% to 2.780%**, while reducing Sharpe from **0.649 to 0.567** and deepening MDD from $-3.693\%$ **to** $-6.415\%$. This was *not* an exogenous market stress test. C_v1/C_v2 were retained as development records, and their limited participation motivated the subsequent Core–Satellite design. (G9-C; G9-FINAL)

# 6 Selecting the Core–Satellite Candidate

## 6.1 From limited participation to a different portfolio architecture

The matched 2010–2016 A/B/C experiments in Section 5 made the exposure trade-off visible: mean realized gross exposure was 51.9% for B_v2, 46.3% for C_v1, and 48.9% for C_v2, compared with approximately 100% for the 60/40 benchmark. Their low drawdowns therefore cannot be interpreted independently of limited market participation. Rather than amplify every agent position mechanically, the team proposed a persistent equity component together with an actively selected satellite. The resulting **Core35–Satellite65** was a *candidate chosen during development*, not the proven optimal split from an exhaustive 35/65 ratio search. (G9-C; G9-FINAL)

Let $e_i$ denote the unit vector for ETF $i \in \{\text{SPY}, \text{IEF}, \text{GLD}, \text{SHY}\}$. For the selected asset $a_t$ and its signed direction $s_{a_t,t} \in \{-1, +1\}$, the weekly target portfolio is

$$\boldsymbol{w}_t^{\text{target}} = 0.35\, \boldsymbol{e}_{\text{SPY}} + 0.65\, s_{a_t,t} \boldsymbol{e}_{a_t}. \tag{6.1}$$

If no instrument is eligible, the satellite is flat: the target remains +35% SPY with 65% unremunerated cash. Confidence determines *eligibility and selection*, not a continuously scaled satellite weight. The rule constrains *target* gross exposure to at most 100%; realized exposures subsequently drift with market prices. (G9-FINAL; G9-CODE)

## 6.2 The frozen satellite-selection rule

At every weekly decision $t$, the selector receives each ETF's Orchestrator direction $s_{i,t}$ and confidence $c_{i,t}$. A candidate must satisfy $s_{i,t} \neq 0$ and $c_{i,t} \geq 0.35$. Eligible assets receive a signed score. Before the SPY preference and hysteresis are applied, the best candidate is

$$q_{i,t} = \frac{s_{i,t} c_{i,t}}{\sigma_{i,t} + 10^{-8}}, \qquad a_t^{\text{raw}} = \arg \max_{i \text{ eligible}} |q_{i,t}|. \tag{6.2}$$

The archived signal files contain `signal` and `confidence`, but *no* volatility column; hence $\sigma_{i,t} = 0.10$ is the **same fallback for every ETF**. The final cross-asset ranking therefore does not compare ETF-specific realized volatility estimates. The selection also implements two explicit rules:

- **SPY trend preference.** If SPY is above its 30-observation *weekly* SMA, a short-SPY signal is blocked; an eligible SPY score receives the multiplier $\min\{2.5, \max\{1.2, P_t/\text{SMA}_{30,t}\}\}$.
- **Switching hysteresis.** The previously selected *asset*, if still eligible, is retained unless the best competing absolute score is at least $1.015$ times its current score. This reduces some asset changes; it does not prevent a sign reversal within the same asset.

The historical tie-order is SHY, IEF, GLD, SPY. These are *implemented rules*, not general features of Core–Satellite investing. (G9-FINAL; G9-CODE)

Table 6.1: *Nominal target examples under the frozen Core–Satellite policy.*

| Satellite selection | Core and satellite | Combined target |
|---|---|---|
| Long SPY | +35% SPY; +65% SPY | +100% SPY |
| Short SPY | +35% SPY; $-65\%$ SPY | $-30$**% net SPY** |
| Long IEF | +35% SPY; +65% IEF | Two-ETF long allocation |
| Short GLD | +35% SPY; $-65\%$ GLD | Equity long / gold short |
| No eligible satellite | +35% SPY; satellite flat | +35% SPY; 65% cash |

The **short-SPY case** is particularly important: the 35% core is not a guaranteed minimum positive equity exposure. It can be offset by the satellite, producing a net $-30\%$ SPY target with a smaller gross directional exposure after netting. (G9-FINAL)

## 6.3 Development selection was a trade-off, not a Sharpe maximum

Table 6.2 reports the archived **2010–2021 development comparison** for the proposed strategy and its two principal benchmarks, using a common daily execution/accounting approach. It must not be combined with the distinct 2010–2016 Level C table or with the final 2022–2026 OOS results.

Table 6.2: *Development comparison, 2010–2021; net simulated returns.*

| Strategy | CAGR (%) | Vol. (%) | Sharpe | MDD (%) |
|---|---|---|---|---|
| Core35–Satellite65 | 12.064 | 13.210 | 0.929 | −18.291 |
| SPY Buy & Hold | 14.767 | 16.948 | 0.898 | −33.717 |
| 60/40 SPY–IEF, weekly | 10.859 | 9.356 | 1.149 | −19.490 |

The candidate offered a higher development CAGR than 60/40 and a marginally shallower maximum drawdown, but **60/40 achieved the higher Sharpe (1.149 versus 0.929)**; passive SPY had the higher

CAGR. Thus the candidate was preserved as a research trade-off combining participation and dynamic selection, *not* as the argmax of development Sharpe. Because the design, agent aggregation and final portfolio choice were influenced by this period, development performance remains **in-sample with respect to final strategy selection**. The frozen evaluation in Section 8 is the relevant subsequent test. (G9-FINAL)

# 7 Frozen Protocol and Execution Engine

## 7.1 Frozen inference and the final test boundary

The selected candidate used information through **31 December 2021** to estimate models and fix agent/Orchestrator settings. The release freeze records SHA-256 digests of **47 code, parameter, model and signal files**; a separate release index covers 13 additional delivery artifacts. Four Random Forest and four SVM model binaries, ETF-specific technical/Orchestrator parameters and the core selector settings were retained for future inference. FinBERT was excluded from this final configuration. The archived OOS workflow validates artifact hashes and does **not** refit models or retune portfolio parameters using 2022–2026 data. These controls preserve the identity of the evaluated candidate; they are not themselves a statistical performance test. (G9-FREEZE; G9-FINAL; G9-CODE)

## 7.2 A close-to-next-open trading timeline

Weekly decisions are made *after* the last available trading close of the week; the associated target can be filled only at the next available trading open. The shared simulator then values positions at *each daily close*. This operational convention differs materially from the earlier Treasury phase's approximate weekly close-to-close implementation.




Figure 7.1: *Final evaluation protocol: a signal computed from the weekly close is not executed at that same close.*

The final common input contained **1,197 raw trading sessions** from 3 January 2022 through 9 October 2026. After starting on the first eligible weekly decision (7 January 2022), the archived record comprises **249 weekly decisions**, but only **248 scheduled rebalances** with a subsequent available open. The final 9 October signal has no later open in the evaluation horizon. The accounting record has **1,193 daily NAV observations** and the core strategy has **151 sessions with nonzero executed trading**. These are distinct counts, not interchangeable measures of trading activity. (G9-FREEZE; G9-OOS)

## 7.3 Self-financing accounting and execution assumptions

The engine holds fractional share quantities $q_{i,t}$ and a cash ledger $C_t$. Let $p_i^{\text{open}}$ be the executable adjusted-open quote and let $\boldsymbol{w}$ be the new signed target. At a rebalance the post-cost NAV $V^+$ is solved from the self-financing equation

$$V^+ + \kappa \sum_i \left| V^+ w_i - q_i p_i^{\text{open}} \right| = V^-, \qquad \kappa = \frac{2+1}{10{,}000}, \tag{7.1}$$

where $V^-$ is pre-trade NAV. New quantities are $q_i^+ = V^+ w_i / p_i^{\text{open}}$. Daily marks satisfy $V_t = C_t + \sum_i q_{i,t} p_{i,t}^{\text{close}}$ after applicable expenses. The target *gross* bound is $\sum_i |w_i| \le 1$; actual daily gross and net exposures can depart slightly from targets through mark-to-market changes. (G9-CODE)

Table 7.1: *Common final execution and measurement assumptions.*

| Dimension | Frozen evaluation convention |
|---|---|
| Initial capital / benchmark | USD 100,000 each; SPY single purchase; 60/40 SPY–IEF weekly targets |
| Commission / slippage | 2 bps + 1 bp per unit of executed traded notional |
| Short-borrow / cash | 1% annual borrow fee accrued on short exposure (actual calendar days / 365); cash return 0 |
| Shares / prices | Fractional positions; adjusted ETF Open for fills and adjusted Close for daily NAV |
| Portfolio and market assumptions | Target gross $\leq 100\%$; simplified short collateral; no dynamic spread, market impact or guaranteed borrow availability |
| Performance convention | 252 trading observations per year; daily risk-free rate set to zero for diagnostic Sharpe |

### 7.4 Comparable accounting, distinct portfolio policies

The same accounting engine evaluates **Core35–Satellite65**, **SPY Buy & Hold** and **60/40 SPY–IEF**; the strategies nevertheless have different intended turnover. SPY executes its initial buy and holds, whereas 60/40 targets 60/40 weekly and the satellite can switch asset and direction. The frozen record reports *nonzero trading sessions*, *scheduled rebalances*, *traded notional turnover* and *borrow fees* separately. All reported metrics are net of modeled costs; these are simulated, not live broker fills. (G9-FINAL; G9-OOS)

## 8 Final Out-of-Sample Evaluation

### 8.1 Main risk–return comparison

The archived frozen test starts with a USD 100,000 pre-execution NAV on **7 January 2022**, first executes at the **10 January** open, and ends on **9 October 2026**. All strategies use the same 1,193 daily NAV dates and the cost/accounting convention in Section 7. The risk-adjusted outcome is reported *relative to two simple investable portfolios*, not merely relative to zero return. (G9-OOS; G9-FREEZE)

Table 8.1: *Frozen final OOS performance, 7 Jan 2022–9 Oct 2026 (net simulation).*

| Portfolio | CAGR (%) | Vol. (%) | Sharpe | MDD (%) |
|---|---|---|---|---|
| Core35–Satellite65 | 5.872 | 11.566 | 0.551 | −26.543 |
| SPY Buy & Hold | 13.116 | 17.294 | 0.799 | −23.424 |
| 60/40 SPY–IEF, weekly | 7.388 | 11.174 | 0.694 | −19.840 |

The candidate's **31.013% cumulative return** translates to **USD 131,013.46** ending NAV. SPY finishes at USD 179,222.59 (79.223% cumulative), while 60/40 finishes at USD 140,133.46 (40.133% cumulative). Sharpe is $\sqrt{252}\,\bar{r}_d/\sigma_d$, where $\sigma_d$ is the sample standard deviation of daily portfolio returns and the **diagnostic risk-free rate is zero**. *Cumulative return and CAGR are different measures.* (G9-OOS; G9-FINAL)

Figure 8.1 shows that Core–Satellite lagged both benchmarks. Its volatility was lower than SPY but not 60/40, while its drawdown was deeper than either. The favorable development profile did not persist in OOS; this result is preserved without post-test retuning. (G9-OOS; G9-FINAL)

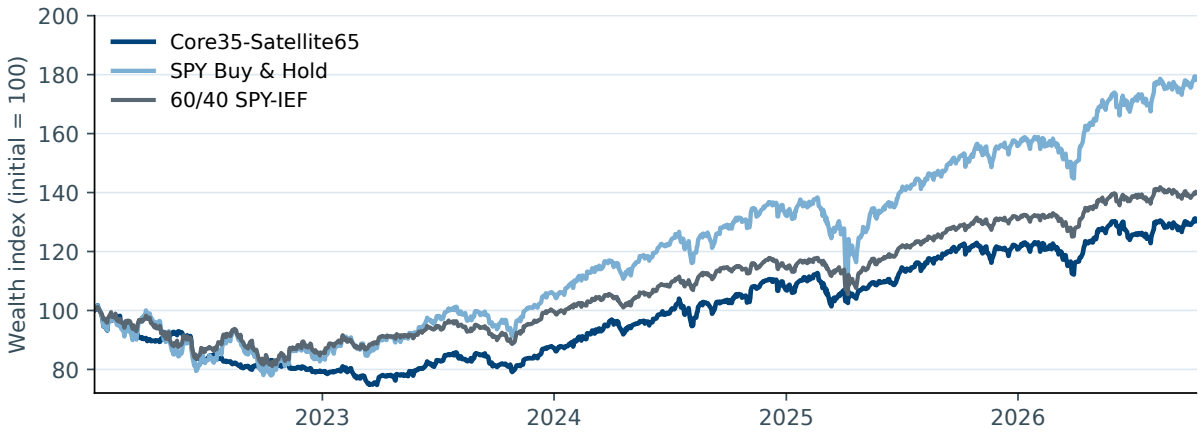



Figure 8.1: *Indexed net daily NAV (7 January 2022 = 100), reconstructed directly from the three archived daily CSVs.*

Table 8.2: *Calendar-year returns (%); 2022 starts 7 January and 2026 ends 9 October.*

| Year | Core35–Satellite65 | SPY Buy & Hold | 60/40 SPY–IEF |
|---|---|---|---|
| 2022 (partial) | −20.79 | −16.03 | −14.48 |
| 2023 | 10.60 | 26.18 | 16.90 |
| 2024 | 22.17 | 24.89 | 14.35 |
| 2025 | 12.78 | 17.72 | 14.52 |
| 2026 (YTD) | 8.53 | 15.06 | 7.05 |

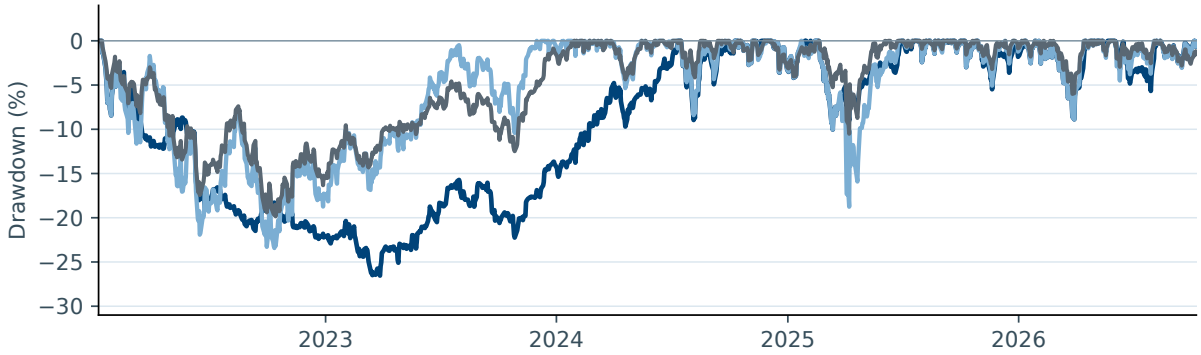



Figure 8.2: *Percentage drawdown from the previous running maximum of each strategy's archived daily NAV.*

## 8.2 Calendar-year returns and drawdown evolution

Core–Satellite lost 20.79% in the partial 2022 sample and captured materially less of the SPY rebound in 2023; its deepest drawdown reached $-26.543$**% on 28 March 2023**. The SPY and 60/40 troughs occurred in October 2022. Accordingly, lower annualized volatility did *not* yield smaller peak-to-trough losses for the candidate (Figure 8.2). The measured association does not isolate whether regime timing, signal aggregation, concentration or transaction costs caused the gap. Those diagnostics are reserved for the subsequent diagnostic chapter. (G9-FINAL; G9-OOS)

**Final assessment.** The protected OOS outcome did not support the initial claim of a superior risk–return trade-off: Core35–Satellite65 underperformed *both* passive comparators on CAGR, Sharpe and maximum drawdown. No parameters or historical observations were changed to improve this recorded outcome.

## 8.3 Temporal robustness: a retrospective 2023–2026 slice

To distinguish the initial adverse year from the remaining realized path, we evaluated the *unchanged archived daily NAV series* over 3 January 2023–9 October 2026. The three portfolio series were rebased to the common value of 100 at the end of December 2022. This is a **retrospective subset of the original OOS run**: there is no re-estimation of RF/SVM, no update of agent or portfolio-selection parameters, and no reset of trading positions. Because the interval was examined after the full result was available, it must not be described as a new independent holdout.

Table 8.3: *Official OOS versus observed retrospective subperiod; the 2023–2026 rows reuse the same NAV path.*

| Strategy | Observation window | CAGR (%) | Sharpe | MDD (%) |
|---|---|---|---|---|
| Core35–Satellite65 | Official 2022–26 | 5.872 | 0.551 | −26.543 |
| | Retrospective 2023–26 | 14.343 | 1.234 | −10.035 |
| SPY Buy & Hold | Official 2022–26 | 13.116 | 0.799 | −23.424 |
| | Retrospective 2023–26 | 22.381 | 1.431 | −18.755 |
| 60/40 SPY–IEF | Official 2022–26 | 7.388 | 0.694 | −19.840 |
| | Retrospective 2023–26 | 14.061 | 1.428 | −10.505 |

The candidate lost **20.787% cumulatively in the partial 2022 OOS**, versus $-16.032\%$ for SPY and $-14.482\%$ for 60/40. Over the subsequent 946 daily NAV observations, its 14.343% CAGR was close to the 60/40's 14.061%, but remained far below SPY's 22.381%. On that restricted subperiod the Core's maximum drawdown was smaller than SPY's ($-10.035\%$ versus $-18.755\%$). The 2022 cumulative loss is not the same statistic as 2022's annualized CAGR ($-21.160\%$ for the Core). These observed differences are time-window diagnostics, not a causal decomposition or a claim that the strategy should have been chosen using a later starting date.

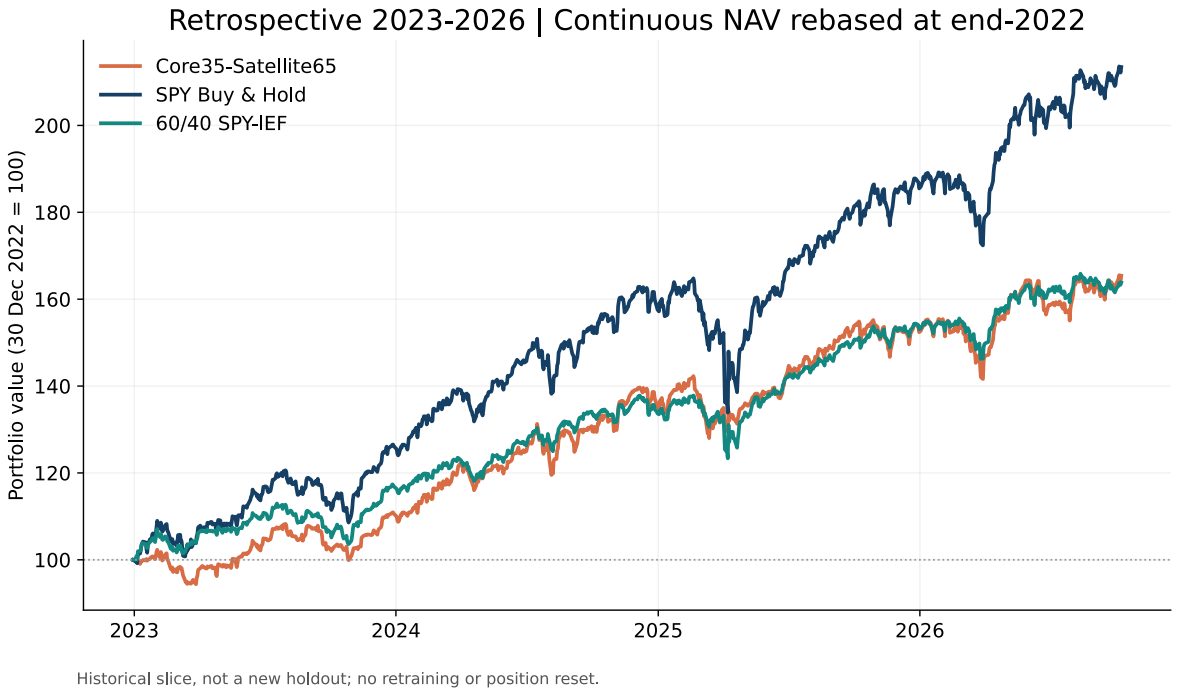



Figure 8.3: *Observed 2023–2026 portion of the frozen 2022-start NAV series, rebased to 100 at the final 2022 NAV. There is no new training, position reset or policy re-selection.*

## 8.4 Transaction-cost ablation on the frozen target calendar

Following the instructor's suggested comparison, a *controlled fixed-policy replay* uses the same frozen target weights, input prices, next-open execution engine and daily accounting as the official test, but sets both trading charges to zero: **2 bps commission → 0 and 1 bp slippage → 0**. The **1% annualized short-borrow fee remains in both runs**; it is not a zero-friction market or a fitted alternative strategy. The cost-inclusive replay is first reconciled to the archived net NAV before the counterfactual is calculated.

Table 8.4: *CAGR sensitivity to trading costs, unchanged 2022–2026 targets (percentage points are arithmetic differences).*

| Strategy | With costs (%) | 0 bps trading (%) | Δ (pp) |
|---|---|---|---|
| Core35–Satellite65 | 5.872 | 6.750 | 0.878 |
| SPY Buy & Hold | 13.116 | 13.123 | 0.007 |
| 60/40 SPY–IEF | 7.388 | 7.408 | 0.021 |

Core–Satellite paid USD 3,883.12 of simulated trading charges and USD 296.52 of short-borrow fees over the historical OOS; the corresponding trading-charge totals for SPY and weekly 60/40 were USD 29.99 and USD 95.28. The Core's higher turnover makes its CAGR more sensitive to trading frictions, but removing these frictions *does not reverse the final relative ranking*: the candidate still trails both benchmarks. The cash charges cannot simply be added to final NAV to obtain a valid alternative CAGR, because trading costs alter subsequent compounding and self-financing position sizes. The reported values come from the replay, not from such an adjustment. The reproducible plotting interface is the repository's `scripts/generate_presentation_charts.py`; its CSV evidence is retained in `tables/` within this companion report.

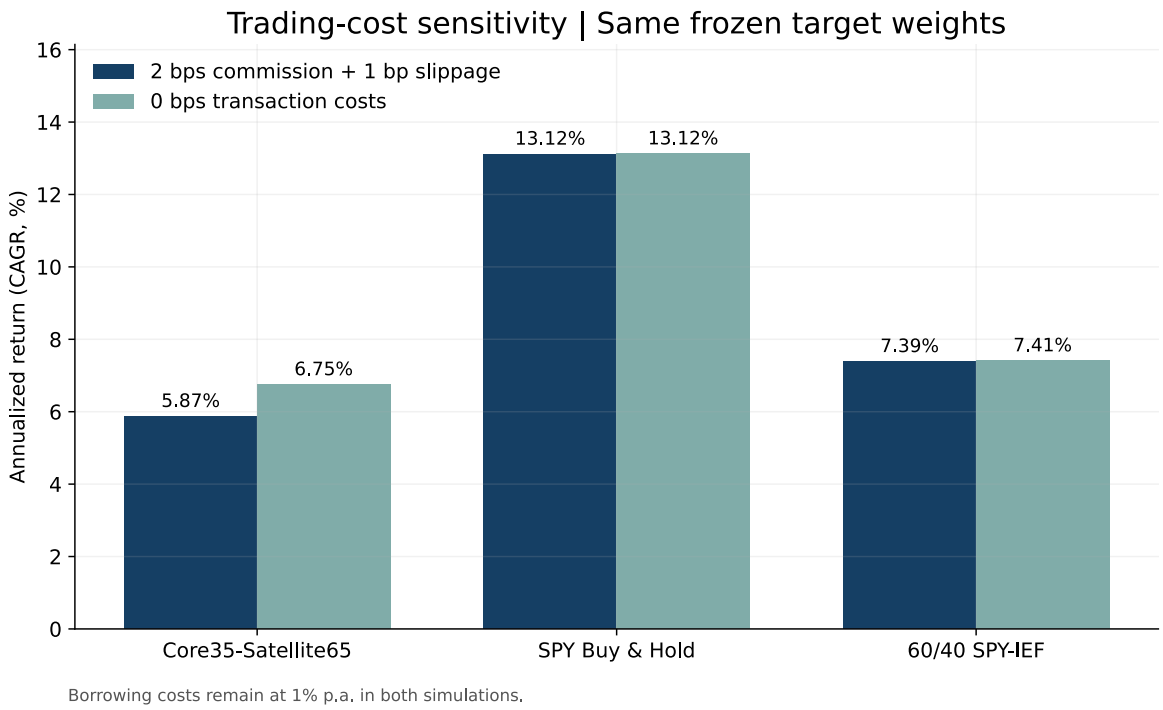



Figure 8.4: *Full OOS trading-cost sensitivity at fixed weights. In the counterfactual, commission and slippage are zero while short borrowing is unchanged.*

# 9 OOS Diagnostics and Economic Interpretation

## 9.1 Realized portfolio concentration

A four-ETF opportunity set does not imply that four assets are held at each decision. The frozen selector assigns the entire nominal 65% satellite to *one* eligible ETF in a long or short direction. Recovering the selected direction from each of the 249 rows in the archived target-weight CSV gives Figure 9.1. Across the final test, SPY was selected in **172 weeks (69.1%)**: 139 long and 33 short. SHY accounted for 54 decisions, IEF for 15 and GLD for 8. The distribution therefore indicates considerable dependence on equity direction even though the candidate was described as multi-asset. (G9-FINAL; G9-OOS)

When SPY was selected short, the nominal $+35\%$ SPY core and $-65\%$ SPY satellite combined to a $-30\%$ **net SPY target**. Hence the 35% core is an architectural component, not a guaranteed minimum long-equity exposure. This result concerns *target weights at decision dates*; exposure between executions changes with relative prices and cashflows. (G9-FINAL)

## 9.2 Target participation and implemented exposure

The target-weight panel has mean gross exposure **90.72%**, compared with **90.60%** for the realized gross-exposure series in the daily portfolio ledger. These are different quantities and sampling frequencies: the former is averaged over 249 decision dates, while the latter is averaged over 1,193 end-of-day NAV observations, including the pre-trade initial date. Their numerical proximity does *not* establish exact execution of every target or identical holdings between rebalances. The strategy still underwent significant directional allocation changes. (G9-OOS; G9-FREEZE)

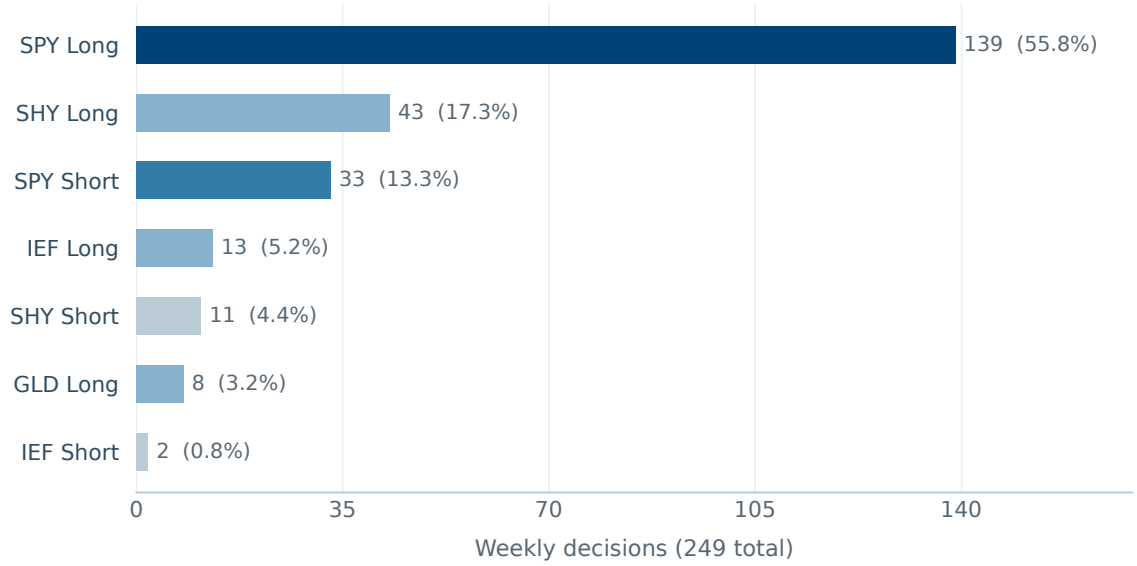



Figure 9.1: *Satellite direction selected at each of 249 weekly decision dates. Counts are reconstructed from the archived* `core_satellite_targets.csv`*; they represent* target decisions*, not executed orders or holding-period returns.*

Table 9.1: *Implementation diagnostics on the common frozen OOS window.*

| Strategy | Trading days | Turnover sum | Trade costs (USD) | Borrow (USD) |
|---|---|---|---|---|
| Core35–Satellite65 | 151 | 130.27 | 3,883.12 | 296.52 |
| SPY Buy & Hold | 1 | 1.00 | 29.99 | 0.00 |
| 60/40 SPY–IEF | 248 | 3.06 | 95.28 | 0.00 |

## 9.3 Trading intensity and implementation costs

A total of **248 scheduled rebalance opportunities** followed the initial decision, but only **151 sessions** generated nonzero trades for the candidate. Turnover 130.27 is the *sum of simulated traded-notional ratios*; it is neither an order count nor an annualized 130.27% rate. By comparison, weekly 60/40 rebalanced on 248 sessions yet incurred much lower turnover: frequency alone does not determine traded notional. (G9-OOS; G9-FINAL)

Transaction costs and borrowing are accounted for in the reported net NAV. The **USD 4,179.64** sum of candidate trading and short-borrow charges represents modeled cash outflows, *not* a valid estimate of the CAGR that would have been earned without those costs. Such a counterfactual is reported as a separate, otherwise identical, fixed-policy replay in Section 8.4. The contribution of concentration, timing, portfolio exposure and implementation costs to relative performance has not been decomposed causally. (G9-FINAL)

**Diagnostic conclusion.** Development-period evidence favored the candidate on selected dimensions, but its frozen OOS performance was inferior to both benchmarks. Strong concentration in SPY and materially higher trading intensity identify mechanisms worth investigating; they do not independently identify a cause of the performance gap.

## 9.4 Exploratory regime adaptation after the official research freeze

A separate research extension evaluates whether the frozen selector can be paired with a simple, observable trend-state switch. At every weekly decision, SPY's close is compared with its rolling **200-trading-session simple moving average**. When SPY is at or above SMA200, the policy follows the original Core35–Satellite65 target. When SPY is below SMA200, it retains the nominal 35% SPY core but restricts Satellite eligibility to ETFs whose **21- and 63-session momentum is positive** *and* whose frozen Orchestrator direction is $+1$. The already implemented confidence threshold, ranking and switching hysteresis are then applied to these eligible agents. Original and defensive selectors maintain independent hysteresis states; the regime flag chooses between their target paths. Signals still use decision-date information and execute next open.

This is *B-style momentum plus positive agent consensus inside a defensive Core–Satellite selector*; it does **not** invoke the complete standalone Level C allocator with its original macro budgets and ex-ante risk cap.

Table 9.2: *Regime-Adaptive SMA200: development and retrospective comparisons at common costs.*

| Strategy | Period | CAGR (%) | Sharpe | MDD (%) |
|---|---|---|---|---|
| Original Core | Development 2010–21 | 12.064 | 0.929 | −18.291 |
| Regime-Adaptive | Development 2010–21 | 10.599 | 0.889 | −16.484 |
| Original Core | Retrospective 2022–26 | 5.872 | 0.551 | −26.543 |
| Regime-Adaptive | Retrospective 2022–26 | 9.805 | 0.912 | −12.802 |
| Defensive always-on | Retrospective 2022–26 | 11.280 | 1.166 | −8.506 |
| 60/40 SPY–IEF | Retrospective 2022–26 | 7.388 | 0.694 | −19.840 |

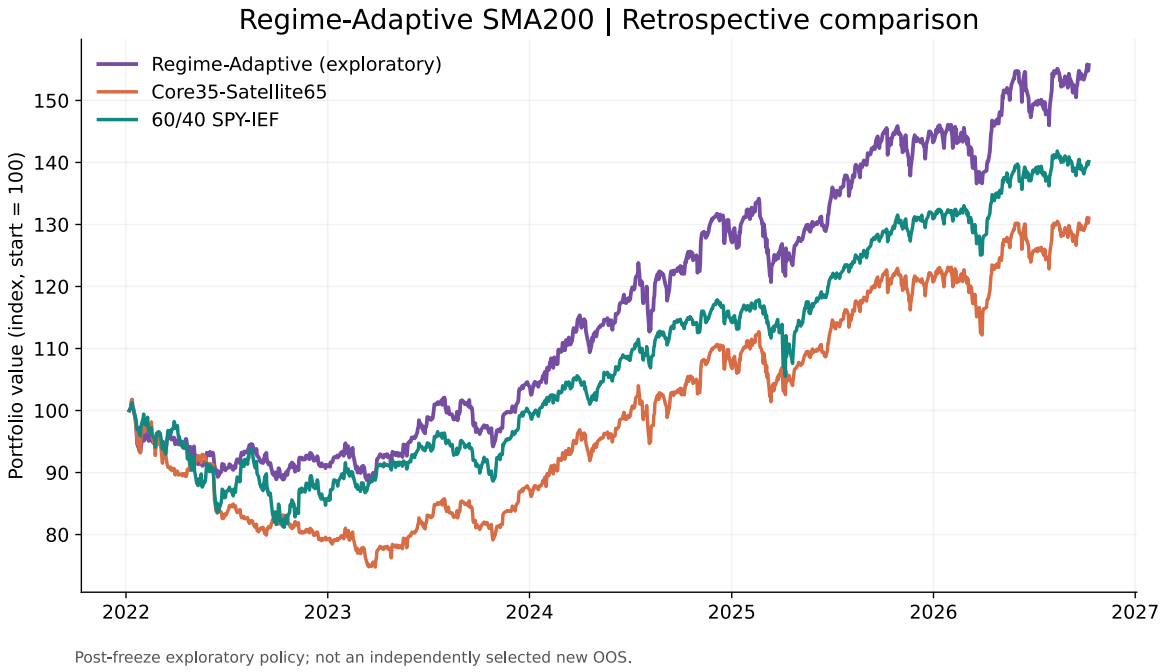



Figure 9.2: *Exploratory 2022–2026 SMA200 regime adaptation with unchanged frozen agent outputs; neither this variant nor its comparator screening constitutes a second independent OOS.*

The below-SMA200 condition occurs at **58 of 249** OOS weekly decisions (and 83 of 626 development decisions). Over the observed 2022–2026 interval, the regime-adaptive policy improves the Core's simulated drawdown and Sharpe, but its development-period CAGR and Sharpe fall relative to the original candidate.

The always-on defensive rule performs better still in this particular retrospective OOS comparison; this is a useful caution against attributing the entire effect to the SMA200 switching mechanism.

**Inference boundary.** The extension was assessed after the official frozen result had been examined. Its direction and moving-average calculations are backward-looking, so there is no identified mechanical look-ahead in signal construction; the scientific concern is instead *post-selection and repeated- comparison bias*. A genuinely new holdout or a prospective monitoring record is necessary before claiming generalized risk improvement or statistically significant alpha. The original official 2022–2026 candidate and its conclusions remain intact.

# 10 Methodological Limitations and Pitfall Analysis

The final OOS experiment evaluates a selected, frozen trading policy under a specified historical information protocol and execution model. General discussions of multiple comparisons and post-selection inference motivate treating repeated development experiments separately from the final holdout (White 2000). It does not turn development performance into independent evidence, nor does a deterministic replay establish statistical superiority. The relevant distinction is between *controls actually implemented* and the *scope of the economic inference*. (G9-FINAL; G9-AB; LUISS-REQ)

## 10.1 Controls and scope of inference

Table 10.1: *Principal research pitfalls, implemented safeguards and remaining scope restrictions.*

| Risk | Implemented control | Interpretation boundary |
|---|---|---|
| Overfitting and model selection | Baselines, logged research steps, frozen models and separate final 2022+ evaluation. | The 2010–2021 comparison was used in strategy selection; it is not an independent performance estimate. |
| Data snooping and repeated comparisons | Historical Treasury/A/B/C trials are retained; the final candidate was not retuned on its OOS outcome. | Earlier development and validation diagnostics have different independence status from the final protected test. |
| Look-ahead and information timing | Rolling features, specified macro availability lags, chronological RF/SVM training and after-close decisions executed next open. | Findings apply to the defined historical input snapshots and source-availability convention. |
| Supervised-model contamination | RF walk-forward predictions; temporal SVM validation with horizon-aware purging. | OOF base-model predictions do not make the GA-selected ensemble OOS. |
| Survivorship and investment universe | Four named, liquid historical ETF proxies; explicit SPY and 60/40 comparators. | This is a fixed-universe ETF study, not a dynamic investable-universe or security-selection test. |
| Text-model information leakage | FinBERT excluded from the final traded policy. | No historical news or sentiment performance contribution is claimed for the frozen candidate. |

## 10.2 Implementation assumptions and generalizability

The portfolio simulator uses auto-adjusted ETF prices, fractional quantities, fixed **2 bps commission + 1 bp slippage** on traded notional, **1% annual short-borrow charge**, unremunerated cash and simplified margin treatment. It does not model dynamic bid–ask spreads, size-dependent market impact, taxes, guaranteed security-borrow availability or operational disruptions. These assumptions make the comparison reproducible; they do not establish implementability at identical prices and fees in a live account. (G9-

FINAL)

The 35/65 allocation was a documented candidate, not the outcome of an exhaustive optimal-ratio sweep. The GA optimizes a *weekly surrogate objective* rather than the full net daily Core–Satellite execution objective. Agent confidence values are model-specific ranking scores, not calibrated probabilities. A nominal 8% ex-ante volatility cap in historical A/B/C configurations is not a promise of 8% realized volatility and does not constrain the eventual Core–Satellite candidate in the same way. (G9-FINAL; G9-AB; G9-C)

Finally, one 2022–2026 historical holdout and three ETF-portfolio comparisons cannot establish statistical alpha or forward profitability. Evidence of mechanism would require a *separately designed* attribution with exposure-matched controls, not a change to the preserved frozen record after inspecting outcomes.

### 10.3 Interpretation of retrospective supplements

The 2023–2026 NAV slice is a descriptive decomposition of the original 2022-start path, not a second independently selected evaluation window. Likewise, the zero-trading-cost experiment is a deterministic execution-cost ablation conditional on fixed target weights; it cannot establish that the underlying signals have predictive alpha. The SMA200 Regime-Adaptive variant was evaluated after the official OOS had been reviewed. Although its trend and agent filters rely only on information available at each weekly decision, the evaluated variant is subject to *post-selection bias*, not automatically to mechanical future-data leakage. None of these supplements replaces the archived 2022-start result, and the observed improvements should not be treated as independently replicated. (G9-OOS; G9-FINAL)

## 11 Software Architecture and Reproducibility

### 11.1 Modular implementation and the release boundary

The research repository contains a modular path from data and feature engineering to agent inference, signal aggregation, portfolio targets, next-open execution and reporting. Figure 11.1 emphasizes functional interfaces rather than a literal Python import graph. Configuration is separate from source code: `config.yaml` preserves the original research rules, while `release_config.yaml` checks the effective final release contract without acting as a tuning interface. (G9-CODE; G9-FINAL)

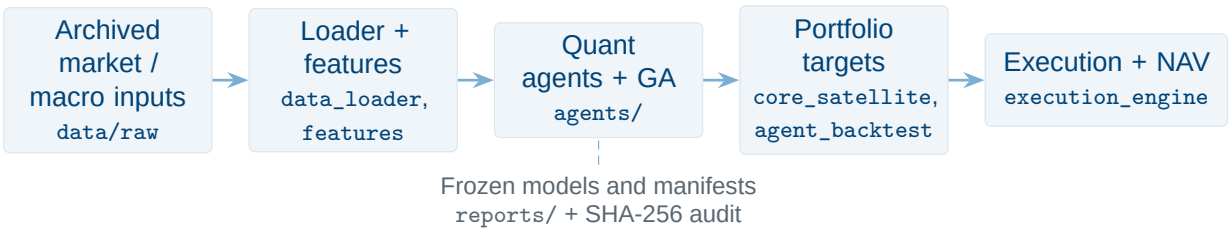



Figure 11.1: *Functional structure of the frozen trading system and supporting integrity layer.*

The historical Git repository additionally retains Treasury research, A/B/C experiment documentation and more extensive testing and CI history. The compact delivered `project/` directory is instead designed for offline review of the frozen selected strategy. It includes raw and processed CSVs, saved RF/SVM binaries, agent signals, evaluation reports and a focused submission test suite, but does *not* reproduce the complete historical research tree. The two artifacts serve distinct purposes and should not be described as identical distributions. (G9-FINAL; G9-AB)

## 11.2 Three distinct integrity layers

Table 11.1: *Verification layers checked in the packaged project snapshot.*

| Control | Files | Purpose |
|---|---|---|
| Scientific development freeze | 47 | Frozen input, code, settings and model digests. |
| Release artifact index | 13 | Links final metrics, signals and provenance to archived hashes. |
| Historical 7 Nov package manifest | 111 | Earlier offline package snapshot; requires regeneration for the eventual final deliverable. |

The earlier 7 November offline package passed its 111-file manifest, all 47 development-freeze checks, 13 release-index checks and the weekly calendar checks, plus six focused submission tests. That historical manifest *must not be reused as the checksum manifest of the final November delivery*: added documentation, presentation and script files require a new package manifest and separate audit. The subsequently expanded local development suite ran **275 tests successfully with five skips**, and GitHub Actions returned successful push/PR checks after correction of source-only CI integration tests. In a source-only checkout, integration tests needing separately distributed frozen OOS artifacts are explicitly skipped; a declared complete release with missing archived inputs should fail. These controls verify software and snapshot integrity, not statistical significance. (G9-FINAL; G9-FREEZE)

## 11.3 Non-destructive replay and reproducible access

The read-only interface displays archived metrics without retraining. Audit mode validates manifests, calendars and source artifacts. Replay mode clones the frozen inputs into a disposable directory, reruns the existing OOS evaluation and compares the original metrics, target weights and three daily ledgers, without overwriting the archived release. In this report build, the audit and packaged test suite were executed; a full OOS replay was *not* required to regenerate the diagnostic figure, which was calculated directly from frozen target rows. (G9-FINAL; G9-FREEZE)

Table 11.2: *Supported offline modes (invoked through the `python main.py` entry point).*

| Mode | Action and scope |
|---|---|
| `audit` | Checks archived checksums, model freeze, and decision calendars; no training. |
| `verify-models` | Loads four frozen GA parameter snapshots with the compatibility adapter. |
| `replay` | Executes the locked OOS backtest in a disposable copy and compares five archived CSV outputs. |
| `CORE development` | Reproduces the selected policy's *development* comparison, not a new final OOS. |
| `--analysis temporal-robustness` | Read-only diagnostics on the archived full and retrospective OOS NAVs. |
| `--analysis regime-adaptive` | Research-only frozen-agent regime replay; not an independent new OOS. |

**Offline review sequence (from `project/`):** `python main.py`; `python main.py --release-action audit`; `python main.py --release-action replay`. The final command is a reproducibility check of the *unchanged* evaluation, not a license to optimize the 2022–2026 observations. Presentation figures and the controlled cost ablation can be reproduced separately using `python scripts/generate_presentation_charts.py --include-adaptive`; its output folder is outside the frozen release records.

# 12 Team Contributions and AI-Assisted Research

## 12.1 Documented areas of work

The project was developed collaboratively by **Gianluca, Giovanni and Gabriele**. Table 12.1 summarizes the main areas of work that can be traced through the research notes, integration record and project history. This is a *contribution map*, not an allocation of exclusive authorship: integrating, reviewing and testing a component also involves work that a single commit cannot describe. (G9-LOG; G9-C; G9-AB)

Table 12.1: *Principal documented contribution areas.*

| Contributor | Main contribution areas |
|---|---|
| Gianluca | Treasury research; A1/A2/B experimental evaluation and Level C comparisons; chronological and accounting audits; research documentation, integration and final release checks. |
| Giovanni | Code refactoring and module boundaries; reintegration of regime clustering; temporal walk-forward SVM replacement; integration and continuous-integration adjustments. |
| Gabriele | Development and testing of technical and machine-learning agents; earlier RF/SVM training and plotting workflows; Orchestrator experimentation and the initial Core–Satellite candidate. |

## 12.2 Role of AI-based development assistance

**ChatGPT** was used to support portions of code inspection, debugging, methodological review, test design, research documentation, interpretation of archived outputs and preparation of communication materials. This is a description of a *support workflow*, not a claim that a language model independently designed, tested or approved the trading strategy. Source code, frozen data, explicit assumptions and the archived evaluation remain the evidential basis for reported results. No additional named AI products or model versions are attributed without a corresponding project record.

The term *agent* has a different meaning inside the system: SMA/EMA, Z-Score and macro rules are quantitative software modules; Random Forest and SVM are trained classifiers; the GA Orchestrator performs rule-based aggregation of their outputs. They are *not* conversational language-model agents. The team remains responsible for the code, the empirical claims and the final interpretation. (G9-CODE; G9-FINAL)

# 13 Conclusion

The research moved from Treasury duration allocation to a four-ETF portfolio comprising SPY, IEF, GLD and SHY. Passive benchmarks, volatility control, deterministic regimes, quantitative agent consensus and Core–Satellite allocation were investigated sequentially. The negative Treasury results, B_v1 ablations and low Level C participation remained part of the recorded experimental process. (G9-TREAS; G9-AB; G9-C)

The frozen evaluation of 7 January 2022–9 October 2026 did **not** support a superior risk–return outcome for the selected Core35–Satellite65 candidate. It delivered **5.872% CAGR**, **0.551 Sharpe** and $-26.543\%$ **maximum drawdown**, compared with 13.116%, 0.799 and $-23.424\%$ for SPY, and 7.388%, 0.694 and $-19.840\%$ for weekly 60/40 SPY–IEF. The development 60/40 portfolio had also shown a higher Sharpe; the final candidate had not been chosen by maximizing that statistic. (G9-OOS; G9-FINAL)

The supplementary diagnostics clarify the result without replacing it. The original Core NAV path, observed from 2023 onward, produced **14.343% CAGR**, close to the weekly 60/40's 14.061% over the same retrospective interval, following a Core cumulative loss of 20.787% during partial 2022. A controlled removal of trading commission and slippage raises full-OOS Core CAGR from 5.872% to **6.750%**, but still does not beat the benchmarks. A motivated SMA200 regime extension produces 9.805% CAGR and $-12.802\%$ maximum drawdown in the 2022–2026 retrospective evaluation; it is research evidence with post-selection uncertainty, not a new independent holdout or proof of alpha.

The principal contribution is an **inspectable, multi-agent research and software workflow**: chronological data handling, distinct A/B/C and ensemble policies, frozen models, next-open execution, cost-aware daily accounting, traceable research decisions and automated verification. The evidence is informative precisely because both positive diagnostics and an unfavorable official comparison have been preserved. Future work should test a separately specified mechanism on genuinely new data and assess exposure-matched controls. The frozen official finding must not be revised to match the strongest retrospective variant. (G9-FREEZE; G9-FINAL)

# References

Academic works provide methodological context, not validation of the project backtests. G9 labels identify distinct research records and frozen artifacts cited in the text.

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## Data and software sources


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## Project archive and course materials


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– (2026). *Frozen Core–Satellite Strategy and Final Evaluation*. [G9-FINAL]. Final scientific protocol; source at commit 437351f.

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# A Experiment and Decision Register

The register summarizes *distinct research decisions*, not every commit or unlogged local run. Each entry includes a hypothesis (H), alteration (M), observation window (P), comparator (B), evidence (R), interpretation/decision (I/D), principal boundary (L) and source (S). The historical Treasury weekly close-to-close records and the later daily next-open multi-asset simulator are distinct measurement systems; their metrics must not be pooled. (G9-TREAS; G9-AB; G9-LOG; G9-C; G9-FINAL)

Table A.1: *Research decisions, including negative and abandoned variants.*

| ID | Hypothesis, change, calendar and benchmark | Result, interpretation, decision, limitation and source |
|---|---|---|
| T1 | **H/M:** Macro scoring can time Treasury duration; original 2Y trend, CPI and curve components, thresholds $\pm 2$. **P:** train 2005–2016; historical validation 2017–2021. **B:** SHY/IEF/TLT buy-and-hold. | **R:** cumulative return $-3.11\%$ train, $+2.72\%$ validation; TLT $+93.50\%$ and $+38.18\%$. **I/D:** retain as negative historical baseline. **L:** different duration risk, previously examined validation. **S:** G9-TREAS. |
| T2 | **H/M:** Wider regime thresholds reduce switches; change $\pm 2$ to $\pm 3$. **P:** 2005–2016 train; 2017–2021 inspected validation. **B:** T1 and passive Treasury ETFs. | **R:** $+34.55\%$ train, $+5.32\%$ validation cumulative. **I/D:** not promoted. **L:** variant explored after observing performance; no passive-validation advantage. **S:** G9-TREAS. |
| T3 | **H/M:** A 30Y yield proxy aligns better with TLT duration than 2Y/10Y. **P:** 2005–2016 train only. **B:** identical score using 2Y, 10Y, 30Y. | **R:** cumulative $-3.11\%$, $+19.20\%$, $+46.96\%$ respectively. **I/D:** explore 30Y. **L:** ex-post proxy choice, unequal duration exposure. **S:** G9-TREAS. |
| T4 | **H/M:** Different defensive legs improve bearish-duration outcomes. **P:** train only. **B:** $+50\%$ SHY/$-50\%$ TLT, 100% SHY, $-100\%$ TLT. | **R:** cumulative $+46.96\%$, $+58.20\%$, $+35.10\%$. **I/D:** no defensive variant promoted. **L:** not equal-risk; shorts have borrowing/financing requirements. **S:** G9-TREAS. |
| T5 | **H/M:** Ridge predicts next-week TLT minus SHY better than an expanding mean. **P:** 463 walk-forward predictions, 2008–2016. **B:** expanding historical-mean predictor. | **R:** RMSE 200.360 vs 199.103 bps; accuracy 50.54% vs 54.43%. **I/D:** reject this specification. **L:** forecast loss, not execution performance; local-only script. **S:** G9-TREAS. |
| T6 | **H/M:** Falling 30Y yield over 13 weeks favors TLT, otherwise SHY. **P:** 2008–2016 train diagnostic; 2017–2021 validation. **B:** matched TLT B&H. | **R:** cumulative $+91.80\%$ vs $+73.19\%$ train; $+7.15\%$ vs $+38.18\%$ validation. **I/D:** pause Treasury tuning. **L:** archived close-to-close, not daily next-open; nonindependent exploration. **S:** G9-TREAS. |
| A1 | **H/M:** Establish investable passive references: SPY, weekly 60/40, equal weight rebalanced/B&H. **P:** training 2005–2016. **B:** each other. | **R:** SPY 7.77% CAGR, 60/40 7.13%, EW weekly 6.54%. **I/D:** retain benchmark ladder. **L:** differing volatility, rebalancing frequency and asset mix. **S:** G9-AB. |
| A2 | **H/M:** Add 126-session covariance and 8% ex-ante vol control to long-only full-investment equal weight. **P:** 2005–2016. **B:** weekly EW. | **R:** Sharpe 0.977 vs 0.942; CAGR 6.24% vs 6.54%. **I/D:** keep risk-control comparator. **L:** 8% estimated target, not realized guarantee or directional forecast. **S:** G9-AB. |

*Table A.1 continued*

| ID | Research setup | Evidence and disposition |
|---|---|---|
| B1 | **H/M:** Combine four VIX/inflation states, 21/63-day momentum, macro gate and short positions. **P:** 2005–2016. **B:** A2/A1. | **R:** B_v1 CAGR 0.24%, Sharpe 0.071, MDD $-17.55\%$. **I/D:** retain as negative ablation baseline. **L:** borrowed-stock and turnover costs, variable exposure. **S:** G9-AB; G9-LOG. |
| B2 | **H/M:** Remove shorts and additional directional macro gate; keep regimes and long/flat momentum. **P:** 2005–2016 train. **B:** B_v1, fixed-budget ablations. | **R:** B_v2 CAGR 2.64%, Sharpe 0.646 vs B_v1 0.071; budget-regime Sharpe 0.6464 vs fixed-mean 0.6094. **I/D:** freeze B_v2, retain B_v1. **L:** development selection and lower gross participation. **S:** G9-AB. |
| B3 | **H/M:** Inspect frozen B_v2 beyond train without modifying rules. **P:** historical validation 2017–2021. **B:** selected B_v2 train and passive references. | **R:** validation CAGR 1.725%, Sharpe 0.5239, MDD $-4.696\%$ (1,259 sessions). **I/D:** preserve diagnostic; no later B retuning. **L:** validation had been consulted, not independent for all later research. **S:** G9-AB; G9-LOG. |
| C1 | **H/M:** Technical/macro/ML consensus can filter B_v2; C_v1 long/flat, C_v2 long/short/flat. **P:** matched 2010–2016. **B:** B_v2 and 60/40. | **R:** Sharpe B_v2 0.393, C_v1 0.607, C_v2 0.649, 60/40 1.143. **I/D:** retain C as development evidence. **L:** OOF base signals do not give OOS GA selection. **S:** G9-C. |
| C2 | **H/M:** Increase allocation scale under 8/12% caps; no new signals. **P:** matched 2010–2016 development. **B:** unscaled C_v2. | **R:** at 12% cap, $1\times \rightarrow 2\times$ raises CAGR 2.092% to 2.780% but reduces Sharpe 0.649 to 0.567. **I/D:** no automatic promotion. **L:** exposure sensitivity, not a market shock stress test. **S:** G9-C. |
| C3 | **H/M:** Replace nearly always-long early SVM with horizon-purged walk-forward 4-week classifier. **P:** internal 2010–2016. **B:** prior SVM. | **R:** selected historical SVM became less uniformly long; inner CV balanced accuracy generally around 0.50–0.59. **I/D:** retain revision for temporal rigor. **L:** a more defensible protocol need not improve returns. **S:** G9-LOG. |
| CORE | **H/M:** Restore participation with 35% SPY core and 65% agent-ranked satellite. **P:** selection development 2010–2021. **B:** SPY and 60/40. | **R:** 12.064% CAGR, 0.929 Sharpe, $-18.291\%$ MDD; 60/40 Sharpe 1.149. **I/D:** select candidate and freeze, *not* as maximum-Sharpe. **L:** 35/65 not proven optimal. **S:** G9-FINAL. |
| OOS | **H/M:** Apply unchanged frozen Core to protected final test with next-open execution. **P:** 7 Jan 2022–9 Oct 2026. **B:** SPY, weekly 60/40. | **R:** CAGR 5.872% vs 13.116% and 7.388%; Sharpe 0.551 vs 0.799 and 0.694. **I/D:** hypothesis not supported; preserve negative result. **L:** no causal attribution or future-profitability claim. **S:** G9-FINAL; G9-OOS. |

# B Agent Specifications and Model Targets

The tables below reflect the *saved final configurations*, not a new hyperparameter search. The original historical experiments are described in Section 5; the frozen files in `project/src/agents/config/` give the complete serialized settings and trained estimators. Each asset has its own saved model/parameter record. (G9-CODE; G9-FREEZE)

Table B.1: *Final technical-agent settings by ETF (Z-Score mode is momentum for all assets).*

| ETF | SMA window | EMA window | Z window | Z entry threshold |
|---|---|---|---|---|
| SPY | 30 | 20 | 60 | 1.50 |
| IEF | 20 | 5 | 40 | 1.75 |
| GLD | 100 | 15 | 60 | 2.00 |
| SHY | 60 | 20 | 50 | 0.75 |

The rule-based macro agent uses a `curve_10y_2y` slope, changes in the ten-year Treasury yield and annual CPI inflation. Its historical saved file specifies four-observation slope/rate windows and a score threshold of 0.5. Macro features are lagged under the common decision-date availability convention. The retained `target_asset` metadata in the macro module reflects earlier Treasury-era naming and is not itself an additional tradable instrument in the final four-ETF universe.

Table B.2: *Frozen RF settings and SVM selected hyperparameters (asset-specific).*

| ETF | RF depth | RF leaf | RF $p$ gate | Trees | SVM kernel / $C$ / $\gamma$ | CV BA | SVM band |
|---|---|---|---|---|---|---|---|
| SPY | 3 | 20 | 0.53 | 100 | rbf / 10.0 / 0.1 | 0.521 | 0.25 |
| IEF | 3 | 20 | 0.52 | 100 | rbf / 10.0 / 0.01 | 0.545 | 0.25 |
| GLD | 3 | 20 | 0.52 | 100 | linear / 10.0 / scale | 0.558 | 0.25 |
| SHY | 3 | 20 | 0.53 | 100 | rbf / 10.0 / 0.1 | 0.564 | 0.25 |

**Predictive labels and training.** RF targets the *next-week* direction and follows rolling historical fit/prediction blocks; SVM targets the *four-week-ahead* direction using a four-week purged time-series validation gap. The saved SVM records use four CV splits, a minimum of 156 training weeks, and a scalar decision-score normalization. The reported CV balanced accuracy is an *internal model-selection diagnostic*, not a return Sharpe ratio. The RF scores and SVM confidence proxies are not calibrated to a shared probability of trading success. (G9-LOG; G9-CODE)

# C Frozen Selector and Orchestrator Parameters

**Selector contract:** SPY core 0.35, Satellite nominal weight 0.65; eligibility requires direction different from zero and confidence at least 0.35. The selector uses score $s\,c/(\hat{\sigma}+\varepsilon)$ with the archive's common 0.10 volatility fallback, a trend preference for SPY above weekly SMA30, and a 1.5% relative switching hysteresis. Confidence ranks eligible choices rather than scaling the nominal 65% continuously. Equation (6.2) gives the mathematical selection rule. (G9-FINAL; G9-CODE)

Table C.1: *Frozen per-asset GA Orchestrator weights ($\sum_j w_j \simeq 1$).*

| ETF | SMA/EMA | Z-Score | Macro | RF | SVM |
|---|---|---|---|---|---|
| SPY | 0.0000 | 0.0265 | 0.5016 | 0.4717 | 0.0001 |
| IEF | 0.2722 | 0.4701 | 0.2517 | 0.0003 | 0.0057 |
| GLD | 0.3189 | 0.2122 | 0.3691 | 0.0000 | 0.0998 |
| SHY | 0.4931 | 0.0169 | 0.4686 | 0.0000 | 0.0214 |

Table C.2: *Frozen confidence exponents and asymmetric decision thresholds.*

| ETF | $\alpha_{\text{SMA}}$ | $\alpha_Z$ | $\alpha_M$ | $\alpha_{RF}$ | $\alpha_{SVM}$ | $\theta_L$ | $\theta_S$ |
|---|---|---|---|---|---|---|---|
| SPY | 3.000 | 2.699 | 1.462 | 1.665 | 1.547 | 0.050 | 0.850 |
| IEF | 2.304 | 0.171 | 3.000 | 1.777 | 2.183 | 0.050 | 0.850 |
| GLD | 2.136 | 0.110 | 0.123 | 2.266 | 2.991 | 0.850 | 0.207 |
| SHY | 1.673 | 0.100 | 1.797 | 2.898 | 2.842 | 0.051 | 0.850 |

Weights and exponents in the printed tables are rounded; a displayed weight of 0.0000 may represent a very small nonzero archived value. The source JSON provides full precision.

The Orchestrator's GA objective is an historical weekly *surrogate* and must be kept separate from the final daily ledger's next-open accounting. Large or tiny weights in Table C.1 are the *saved solution for each asset*, not evidence that the corresponding predictor was independently causal. The final selected policy uses the archived snapshots unchanged. The machine-readable JSON records are authoritative when greater numerical precision is required. (G9-CODE; G9-FREEZE)

# D Supplementary Performance Tables

Table D.1 preserves all nine matched A/B/C portfolio comparisons on the *2010–2016 training calendar*. It is not the final strategy-selection development comparison and it is not the 2022–2026 final OOS. (G9-C)

Table D.1: *All A/B/C portfolios on the common 2010–2016 calendar.*

| Strategy | CAGR (%) | Vol. (%) | Sharpe | MDD (%) | Mean gross (%) |
|---|---|---|---|---|---|
| SPY Buy & Hold | 12.260 | 15.441 | 0.826 | -18.606 | 99.943 |
| 60/40 weekly | 9.566 | 8.296 | 1.143 | -8.470 | 99.943 |
| Equal weight weekly | 4.688 | 5.802 | 0.819 | -7.571 | 99.943 |
| Equal weight buy-and-hold | 5.031 | 6.076 | 0.838 | -7.678 | 99.943 |
| A2 volatility control | 4.653 | 5.780 | 0.816 | -7.571 | 99.943 |
| B_v1 | -1.073 | 4.925 | -0.194 | -17.553 | 66.680 |
| B_v2 | 1.366 | 3.621 | 0.393 | -7.011 | 51.939 |
| C_v1 | 1.793 | 3.001 | 0.607 | -3.693 | 46.302 |
| C_v2 | 2.091 | 3.272 | 0.649 | -3.693 | 48.864 |

The cost and turnover decomposition appears in Table 9.1; it is intentionally not repeated here. A *trade day* is a session with nonzero executed trading activity, not an order count. The annual OOS return breakdown is reported in Table 8.2. Figures 8.1, 8.2 and 9.1 were generated from the saved daily/target ledgers. (G9-OOS; G9-FINAL)

# E Repository Audit and Source Provenance

The scientific release audit checks 47 frozen files and 13 indexed artifacts. An *earlier* 7 November offline package contained a valid 111-file manifest and six focused submission tests; these counts do not describe the not-yet-assembled final delivery ZIP. The updated development suite completed 275 local tests with five skipped and a successful GitHub CI check on the corrected branch. The final package must receive its own manifest and separate checksum audit before submission. These checks verify snapshot consistency, not investment performance.

Table E.1 identifies the controlling evidence for the main categories of claims. It deliberately distinguishes historical research notes, the complete software history and the frozen submitted evaluation. A citation to a code module provides a specification; a citation to the daily ledger provides an observed simulation result. (G9-FINAL; G9-AB; G9-OOS)

Table E.1: *Evidence and repository map for the companion report.*

| Claim family | Primary evidence | Report location |
|---|---|---|
| Treasury history | `docs/treasury_research_review.md` at `437351f`; original cumulative-return tables and experiment notes. | Sec. 3; App. A |
| A/B and ablations | `docs/research_design.md`, `docs/decision_log.md`; historical train/validation results. | Sec. 4 |
| Level C / ML | `docs/level_c_research.md`; `src/agents/`; saved JSON and binaries. | Sec. 5; Apps. B–C |
| Final portfolio | `src/core_satellite.py`, `src/agent_backtest.py`, `config.yaml`, `reports/development_freeze_2021.json`. | Secs. 6–7 |
| OOS metrics | `reports/oos_final_2022_2026/` daily CSVs, target CSV and `comparison_metrics.csv`. | Secs. 8–9; App. D |
| Supplementary analyses | `scripts/generate_presentation_charts.py`, `src/submission_runner.py`, `src/level_c_strategy.py`; CSV snapshots in this report's `tables/`. | Secs. 8–10 |
| Release audit | Earlier `PACKAGE_MANIFEST.json`, release index, freeze record, Git/CI evidence; new manifest due at final packaging. | Sec. 11; App. E |